Remote Sensing of Environment 341 (2026) 115442

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

Integrating multi-geometry InSAR and explainable machine learning for mapping potential landslide areas in reservoir regions

Ke Xing [a], Suju Li [b], Jie Dou [a,c,*], Fang Chen [d,e,f], Lizhe Wang [g], Aonan Dong [c], Hamza Daud [c], Lele Zhang [c]

[a] School of Future Technology, China University of Geosciences, Wuhan 430074, China
[b] National Disaster Reduction Center of China, Beijing 100124, China
[c] Badong National Observation and Research Station of Geohazards, China University of Geosciences, Wuhan 430074, China
[d] International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China
[e] Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China
[f] University of Chinese Academy of Sciences, Beijing 100049, China
[g] School of Computer Science, China University of Geosciences (Wuhan), Wuhan 430074, China

ARTICLE INFO

Edited by: Jing M. Chen

Keywords:
Landslide susceptibility
Multi-geometry InSAR
Explainable machine learning
Negative sampling
2D deformation
SHAP

ABSTRACT

Reservoir-induced landslides represent a persistent hazard in steep mountainous regions, where complex hydro-mechanical interactions and geomorphic processes challenge reliable hazard assessment. However, remote sensing-based landslide susceptibility mapping (LSM) remains limited by uncertain negative sample selection, one-dimensional (1D) InSAR observations, and weak integration between deformation monitoring and environmental modeling. This study proposes an interpretable remote sensing-driven framework that integrates optimized sampling, multi-geometry InSAR deformation, and explainable machine learning (ML) to improve state-based delineation of potential landslide areas (PLAs) in the Baihetan Reservoir region, southwestern China. Dual-orbit Sentinel-1 SBAS-InSAR time series were decomposed into vertical and horizontal (east-west) components to reconstruct two-dimensional (2D) deformation fields. An information value-based negative sampling strategy was used to enhance model reliability, while a data-driven spatial classifier combined with Shapley additive explanations (SHAP) was employed to quantify the relative contributions of key environmental factors. Integrating susceptibility mapping with InSAR deformation through a contingency matrix increased the area of delineated potential landslides overlapping validated landslides by more than fourfold relative to the conventional threshold method (7.29 km^2 vs. 1.76 km^2). The dual-orbit analysis captured localized vertical subsidence rates of up to 93 mm/yr, effectively delineating active zones near reservoir margins, while the decomposed horizontal component provided additional support for kinematic interpretation. This framework advances InSAR-informed landslide assessment by integrating susceptibility and deformation information. The 2D decomposition further supports kinematic interpretation of the identified instability patterns, offering a practical basis for deformation-informed hazard monitoring in reservoir-affected landscapes.

1. Introduction

Reservoir-induced landslides represent a major environmental hazard in steep mountainous regions, where rapid water-level fluctuations, tectonic activity, and complex lithological conditions interact to destabilize slopes (Agliardi et al., 2020; Hu et al., 2020). Cyclic impoundment and drawdown processes alter pore-water pressure, groundwater circulation, and stress conditions within reservoir banks, often triggering progressive slope deformation and episodic failure. In recent years, advances in Earth observation technologies have significantly improved the capability to monitor such hazards at regional scales (Marmoni et al., 2025). Among these technologies, time-series synthetic aperture radar interferometry (InSAR) has become a key tool for detecting surface deformation and identifying slow-moving landslides that may precede catastrophic collapse (Liu et al., 2024; Yan and Wang, 2021).

Despite these advances, reliably delineating landslide-prone zones in reservoir environments remains challenging. Three critical limitations persist in current remote sensing-based susceptibility assessment

* Corresponding author at: School of Future Technology, China University of Geosciences, Wuhan 430074, China.
E-mail address: doujie@cug.edu.cn (J. Dou).

https://doi.org/10.1016/j.rse.2026.115442
Received 19 December 2025; Received in revised form 11 April 2026; Accepted 20 April 2026
Available online 28 April 2026
0034-4257/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

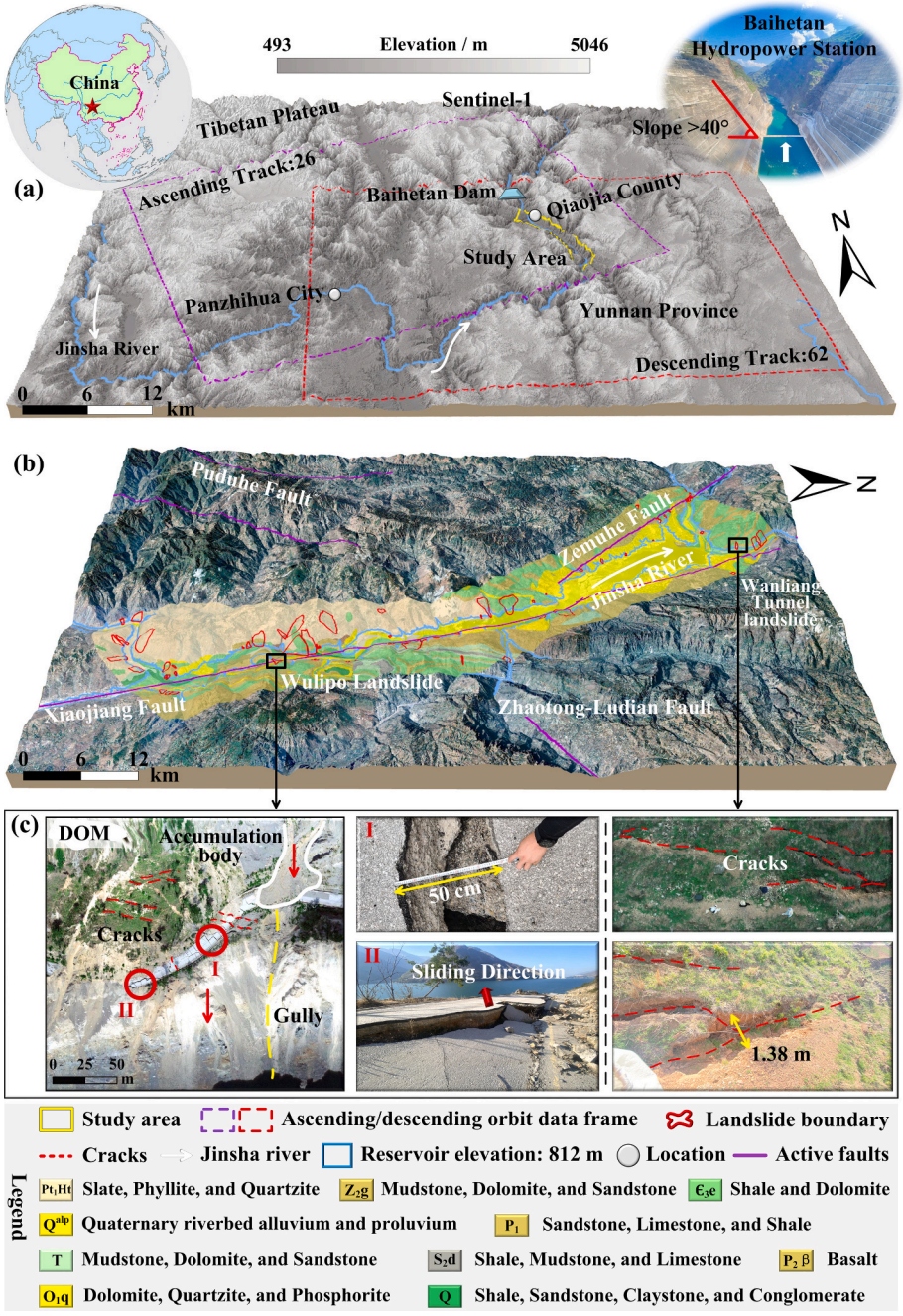

Fig. 1. Overview of the study area. (a) InSAR orbit coverage and geographic location. (b) Active faults, stratigraphic lithology, and landslide locations. (c) Field investigation of the Wulipo and Wanliang Tunnel landslides.

frameworks (Daud et al., 2025; Dou et al., 2020). First, the identification of reliable non-landslide (stable) samples remains problematic (Wang et al., 2025). Most susceptibility studies assume randomly distributed stable samples; however, in reservoir terrains, many slopes exhibit slow or intermittent deformation that may not be visible in optical imagery. Misclassifying such slopes as stable introduces noise into the training dataset and can reduce the predictive reliability and generalization performance of susceptibility models (Guo et al., 2024). Second, conventional InSAR analyses typically rely on single-orbit line-of-sight (LOS) observations, which provide only 1D deformation measurements (Sharifi et al., 2024). In complex mountainous terrain, slope movements frequently occur in multiple directions due to combined gravitational, hydrological, and structural controls. As a result, LOS measurements may not fully capture dominant kinematic components (Kang et al., 2021). Multi-geometry InSAR observations derived from ascending and descending orbits can reconstruct 2D deformation fields, thereby offering a more comprehensive description of slope kinematics. However, such kinematic information is rarely incorporated into landslide susceptibility models (Du et al., 2025). Third, deformation monitoring and susceptibility modeling are typically conducted as independent analyses. Susceptibility models describe long-term environmental predisposition to slope failure, whereas InSAR observations capture the current kinematic state of slopes. The absence of an effective framework integrating these complementary datasets limits the identification of dynamically evolving hazards, particularly in reservoir settings where hydro-mechanical forcing changes continuously through impoundment and drawdown cycles (Tian et al., 2025).

Recent developments in ML have improved landslide susceptibility

modeling by capturing nonlinear relationships among environmental conditioning factors (Dou et al., 2020). However, many models remain difficult to interpret for hazard assessment and decision-making (Reichenbach et al., 2018). Explainable artificial intelligence (XAI) approaches, such as SHAP, can improve model transparency by quantifying the contributions of environmental variables (Zheng et al., 2025). In parallel, existing InSAR-assisted LSM approaches can be broadly grouped into three categories: overlay-based screening, deformation-informed predictors, and multi-temporal updating schemes, all of which seek to better reflect ongoing slope activity (Yao et al., 2023). These strategies offer practical value for integrating surface deformation into susceptibility assessment. In reservoir settings, however, their performance remains constrained by LOS geometry, topographic distortion, temporal decorrelation, and the transient nature of deformation signals, all of which may introduce observational bias (Chen et al., 2023). Moreover, threshold-based fusion is often too rigid to represent gradual variations in hazard state, whereas directly embedding deformation into susceptibility models may blur the distinction between long-term predisposition and short-term kinematic state (Tian et al., 2025). For this reason, InSAR-derived deformation is more appropriately treated as a dynamic state indicator that complements, rather than replaces, susceptibility modeling (Song et al., 2025).

Despite these advances, a conceptual gap remains in current landslide hazard assessment frameworks. Deformation and susceptibility are still commonly handled as separate spatial layers, rather than as complementary indicators of slope instability under different states (Dai et al., 2023). This separation is particularly problematic in reservoir environments, where slopes may remain dormant despite strong geological predisposition, or may enter active deformation under transient hydro-mechanical forcing. As a result, deformation-based screening can miss hazardous but weakly expressed slopes, whereas susceptibility models alone cannot resolve slopes already undergoing active kinematic change. This disconnect limits the reliable delineation of PLAs in dynamically evolving reservoir settings.

To overcome the limitations of existing approaches, this study develops an interpretable remote sensing framework that integrates multi-geometry InSAR observations with ML-based landslide susceptibility modeling for state-based hazard delineation. In this framework, landslide conditions are represented through the integration of static susceptibility levels and observed remote sensing deformation. The framework combines optimized negative sample selection, dual-orbit SBAS-InSAR deformation reconstruction, and SHAP-based model interpretation within a contingency matrix fusion strategy. By conceptualizing landslide hazard as a coupled state defined by susceptibility and deformation intensity, the proposed approach enables a more robust representation of slope instability conditions.

Using the Baihetan Reservoir region in southwestern China as a case study, this research makes three main contributions: (1) a negative sample optimization strategy that reduces noise in susceptibility modeling and improves predictive reliability in deformation-prone reservoir terrains; (2) a multi-geometry InSAR-based kinematic reconstruction framework that resolves 2D deformation patterns and improves the characterization of slope movement processes; (3) a state-based hazard delineation approach based on a contingency matrix, which integrates deformation observations and susceptibility predictions to identify PLAs more effectively than conventional threshold-based methods.

Together, these contributions advance the integration of remote sensing deformation monitoring and susceptibility modeling, providing a physically interpretable framework for landslide hazard assessment in reservoir environments and offering new opportunities for remote sensing-driven geohazard monitoring.

Table 1
Data sources and characteristics used in this study.

Dataset	Source	Data format	Resolution/ Scale
Landslide Inventory Database	Historical Archives and Remote Sensing Interpretation	Vector	
DEM	Copernicus Data Space Ecosystem	Raster	30 m
Geological Datasets	China Geological Survey (CGS) GeoCloud	Vector	1:200,000
Sentinel-1	Alaska Satellite Facility (ASF)	Raster	5 m × 20 m
Vegetation Indices	Geospatial Data Cloud	Raster	30 m
Hydrological Datasets	National Geomatics Center of China (NGCC)	Vector	1:250,000
Road Infrastructure Data	National Geomatics Center of China (NGCC)	Vector	1:250,000
Meteorological Data	China Meteorological Administration (CMA)	Vector	
Orthophoto	UAV	Raster	0.03–0.05 m
LiDAR Data	Airborne LiDAR	Point Cloud	0.05 m

2. Study area and datasets

2.1. Study area

The Baihetan Reservoir area lies along the southeastern margin of the Tibetan Plateau and is characterized by intense Cenozoic tectonic activity and a typical plateau-mountain tectonic framework (Fig. 1a). The Jinsha River traverses this region through steep, V-shaped valleys. Since impoundment began in April 2021, a wide hydro-fluctuation zone has developed, with water levels oscillating between 765 m and 825 m. These periodic drawdown cycles, coupled with continuous water-rock interactions and toe erosion, significantly alter pore-water pressure regimes and degrade the mechanical properties of slope materials, driving hydro-mechanically induced instability along the reservoir banks. Structurally, the area is transected by major near-north-south active faults, notably the Zemuhe and Xiaojiang Faults, which parallel the river (Yao et al., 2025) (Fig. 1b). While the regional stratigraphy is extensive, slope vulnerability is primarily governed by mechanically weak lithological assemblages. Specifically, fractured basalts, alternating soft-hard interbeds (e.g., sandstone-mudstone), and thick Quaternary deposits increase the susceptibility to slope failure (Fig. 1c). These features also pose unique challenges for landslide monitoring and zoning, underscoring the need for integrated, deformation-informed assessment frameworks.

2.2. Data sources

This study employed the multi-source datasets detailed in Table 1. A landslide inventory comprising 74 events (including 12 bank collapses) was mapped from historical archives and field surveys. These positive samples were identified strictly by morphological and historical evidence, independent of current deformation rates. This ensures that the spatial models capture geological predisposition without bias from deformation-related signals. The inventory was partitioned into a modeling dataset (59 landslides, 11.4 km^2; 7:3 train-test split) and an independent external validation set (15 landslides). Environmental factors (Fig. 2) were extracted from a 30-m Copernicus DEM (topography) and 1:200,000-scale maps (geology). For kinematic observations, 103 Sentinel-1 images (from January 9, 2020, to May 29, 2023) were acquired to derive regional surface deformation via time-series InSAR.

3. Methodology

A deformation-informed susceptibility assessment framework was developed by integrating multi-source data, ML, and InSAR-derived

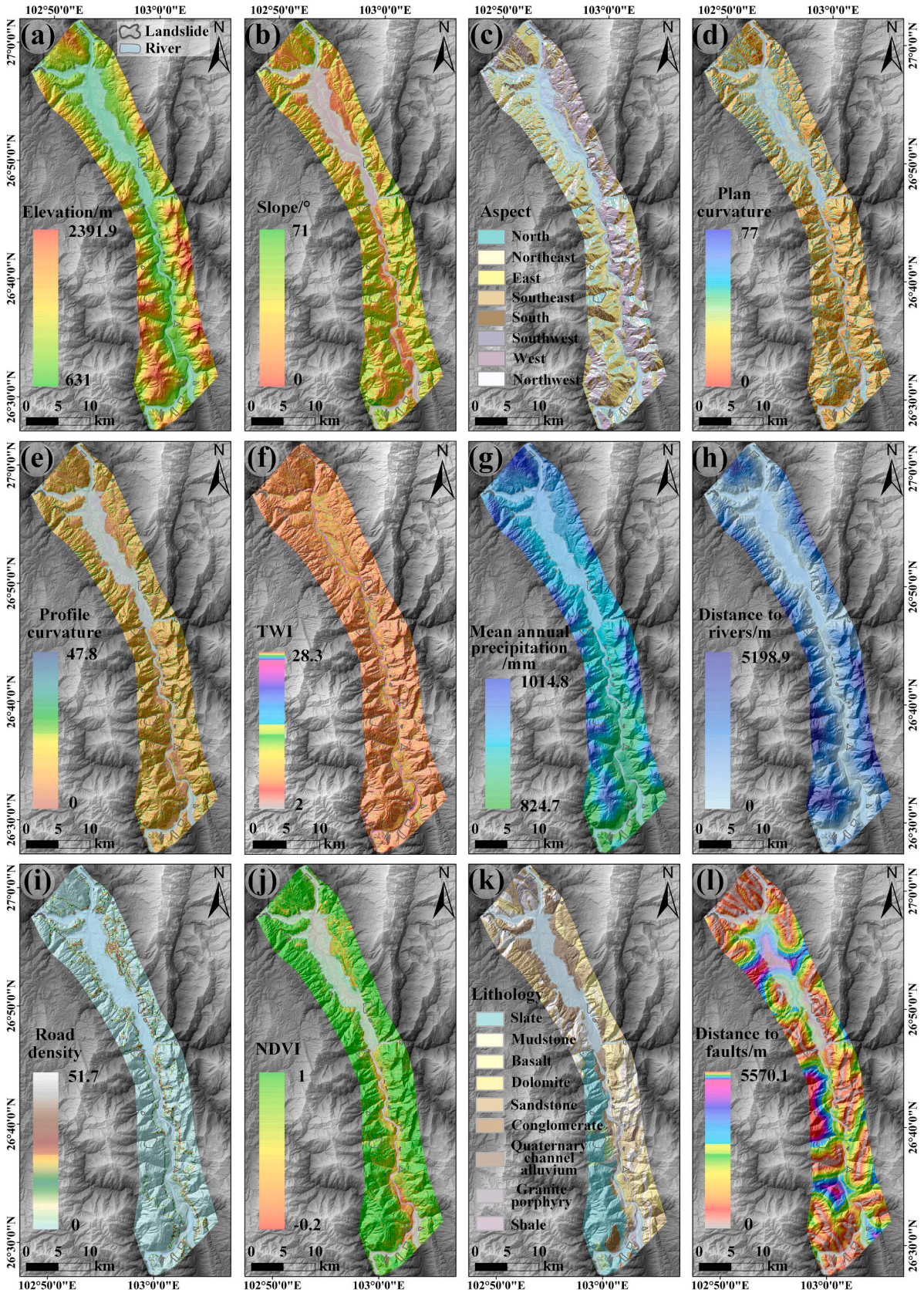

Fig. 2. Landslide conditioning factors. (a) Elevation. (b) Slope. (c) Aspect. (d) Plan curvature (Plan_c). (e) Profile curvature (Profile_c). (f) Topographic Wetness Index (TWI). (g) Mean annual rainfall. (h) Distance to river (Dis_River). (i) Road density (Den_Road). (j) Normalized Difference Vegetation Index (NDVI). (k) Lithology. (l) Distance to fault (Dis_Fault).

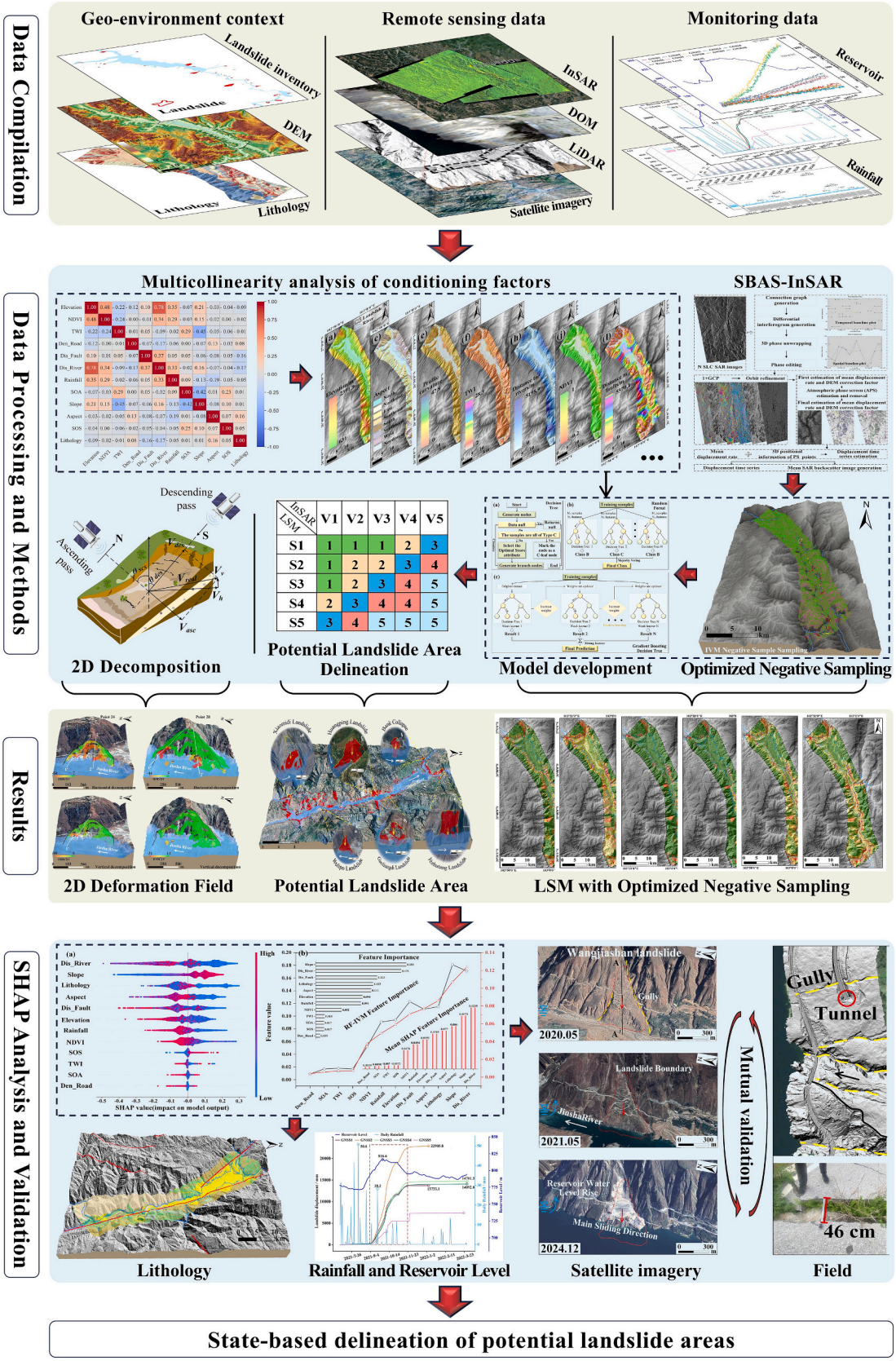

Fig. 3. Methodological flowchart.

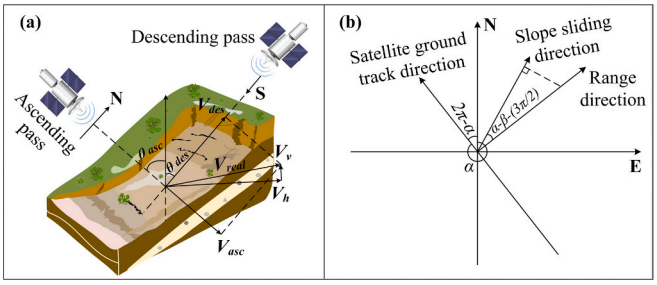

Fig. 4. Geometry of 2D deformation decomposition. (a) Imaging geometry and vector decomposition. (b) Planar relationship between satellite heading and surface displacement vectors.

deformation (Fig. 3). The workflow comprises five core steps: (1) factor collinearity analysis; (2) optimized negative sampling; (3) state-based hazard delineation; (4) 2D kinematic decomposition; and (5) SHAP interpretation and validation.

3.1. SBAS-InSAR

To mitigate severe temporal decorrelation caused by complex mountainous terrain and heterogeneous vegetation (average coherence < 0.8) (Zhang et al., 2024a, 2024b), the SBAS-InSAR approach was selected over PS-InSAR because it exploits distributed scatterers and can maximize coherent pixel density (Berardino et al., 2002). In total, 103 SAR images were processed (Interferometric Wide Swath mode, VV + VH polarization, spatial resolution 5 × 20 m, and an incidence angle of 39.17°). Using temporal and perpendicular-baseline thresholds of 60 days and 200 m, respectively, 310 interferometric pairs were generated. The interferometric phase at any pixel in the j-th interferogram can be expressed as:

$$\frac{N+1}{2} \le M \le N\left(\frac{N+1}{2}\right) \tag{1}$$

$$\delta\varphi^j = \varphi(t_B) - \varphi(t_A) \approx \delta\varphi^j_{def} + \delta\varphi^j_{topo} + \delta\varphi^j_{atm} + \delta\varphi^j_{noise} \tag{2}$$

where N is the total number of SAR images, and M is the number of interferograms. $\delta\varphi^j$ is the interferometric phase, $\varphi(t_A)$ and $\varphi(t_B)$ are the phase values at times t_A and t_B, respectively, with respect to the initial acquisition time t_0; $\delta\varphi^j_{def}$, $\delta\varphi^j_{topo}$, $\delta\varphi^j_{atm}$, $\delta\varphi^j_{noise}$ denote the phase difference induced by displacement phase, topographic error, atmospheric delay, and noise, respectively.

The processing was conducted within the SARScape environment; the standard processing workflow utilized a 30 m DEM for topographic correction, Goldstein filtering for noise reduction, and the Minimum Cost Flow (MCF) algorithm for phase unwrapping. Following orbit refinement and flattening, Singular Value Decomposition (SVD) was applied to separate phase components and retrieve geocoded time-series deformation (Yang et al., 2023).

3.2. 2D decomposition of multi-geometry data

Relying on 1D LOS measurements often underestimates the actual displacement magnitude in reservoir environments, where slope movements are frequently dominated by slope-parallel horizontal components rather than pure vertical subsidence (Herrera et al., 2013). To better characterize these complex kinematic patterns, a 2D decomposition of the dual-orbit datasets was performed, separating the deformation field into vertical and east-west components (Fig. 4). This geometric separation provides useful kinematic constraints for interpreting possible deformation modes, such as subsidence-dominated bank collapse and laterally dominated slope movement.

The relationship between the LOS deformation rate and the decomposed 2D displacement components is given by:

$$V_{asc} = V_V\cos\theta_{asc} - V_h\cos(2\pi - \alpha_{asc})\sin\theta_{asc} \tag{3}$$

$$V_{des} = V_V\cos\theta_{des} - V_h\cos(2\pi - \alpha_{des})\sin\theta_{des} \tag{4}$$

In the model, V_{asc} and V_{des} represent the LOS deformation velocities derived from ascending and descending SAR acquisitions, respectively. V_v and V_h denote the vertical and horizontal components of surface deformation, respectively; here, the horizontal component is primarily represented by east–west motion because the north–south component is

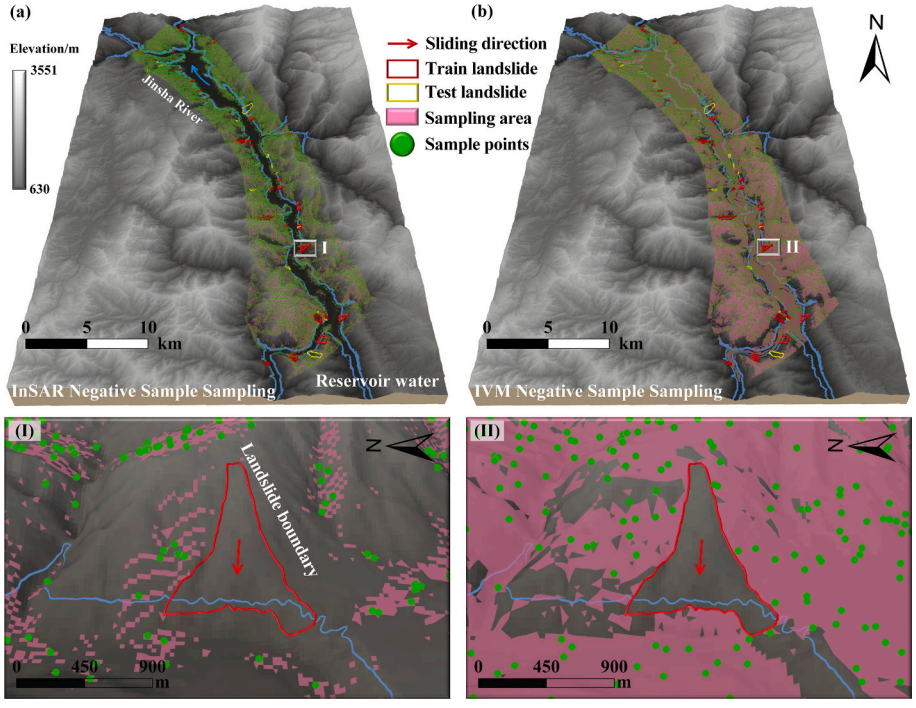

Fig. 5. Negative sampling strategies. (a) InSAR-based sampling. (b) IVM-based sampling. (I-II) Close-up views of selected negative samples.

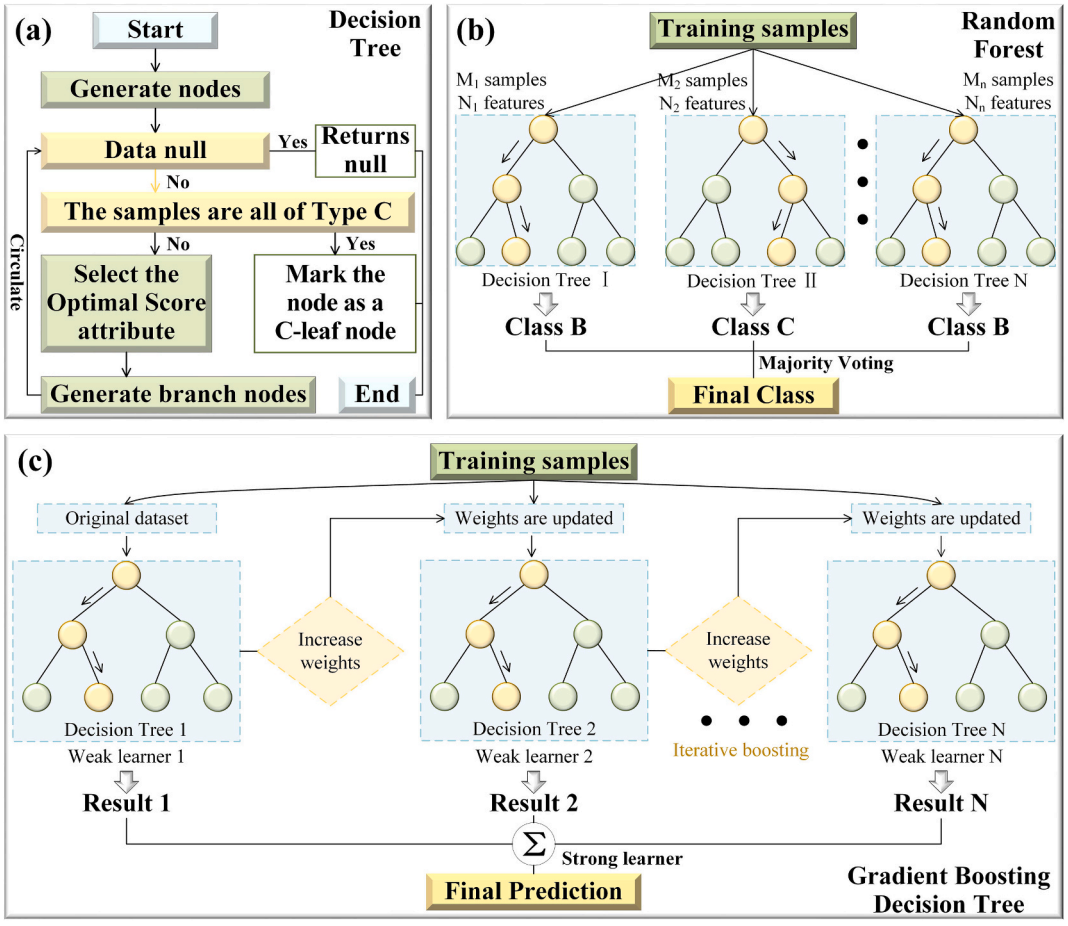

Fig. 6. Tree-based ML models. (a) Decision Tree. (b) Random Forest. (c) Gradient Boosting Decision Tree.

weakly constrained by the SAR viewing geometry. θ_{asc} and θ_{des} refer to the radar incidence angles of the ascending and descending tracks, while α_{asc} and α_{des} indicate the corresponding flight azimuth angles of satellites.

By applying a linear transformation to the LOS deformation velocities from the dual orbits, the vertical and east-west components of the 2D deformation field can be estimated. The solution can be expressed as:

$$V_h = \frac{V_{asc}\cos\theta_{des} - V_{des}\cos\theta_{asc}}{\cos(2\pi - \alpha_{des})\sin\theta_{des}\cos\theta_{asc} - \cos(2\pi - \alpha_{asc})\sin\theta_{asc}\cos\theta_{des}} \quad (5)$$

$$V_v = \frac{V_{asc} + V_h\cos(2\pi - \alpha_{asc})\sin\theta_{asc}}{\cos\theta_{asc}} \quad (6)$$

3.3. Negative sample optimization

Extracting reliable "stable" environmental samples is crucial for minimizing label noise in spatial modeling (Dong et al., 2023). To enhance discrimination of critical predisposing conditions, this study employed two optimization strategies to construct non-landslide control groups at a 1:1 ratio with positive samples:

(1) InSAR-based filtering: InSAR-derived deformation velocities were employed to identify areas exhibiting stable surface conditions (Fig. 5a). Here, InSAR is applied not to define landslides but to filter the negative (stable) samples. By excluding areas with high deformation rates from the non-landslide pool, this strategy helps ensure that the control group represents relatively stable areas, thereby enhancing the model's ability to discriminate the critical environmental conditions governing susceptibility.

(2) IVM-based selection: The IVM, a bivariate statistical technique quantifying factor-landslide associations, was used to generate a preliminary susceptibility map. "Stable" samples were exclusively extracted from areas classified as very low to moderate susceptibility (Fig. 5b), thereby reducing the risk of misclassifying geologically predisposed, yet temporarily dormant slopes as negative samples.

While the proposed strategy effectively reduces label noise by prioritizing sample purity, it introduces a spatial prior that may limit the representativeness of the non-landslide class. Conversely, purely random sampling in reservoir regions risks introducing false negatives, as dormant or intermittently deforming slopes are widespread. For this reason, the present study adopts the optimized strategy as a practical means of reducing label contamination in a deformation-prone reservoir setting, rather than as a universally optimal sampling rule. Hybrid strategies that combine optimized controls with randomly sampled background points warrant further evaluation in future work.

3.4. Susceptibility model construction and evaluation

A suite of tree-based spatial models (Fig. 6), including Decision Tree (DT), RF, and Gradient Boosting Decision Tree (GBDT), was employed (Breiman, 2001; Friedman, 2001). These tree-based models were selected because they perform well on structured tabular data and can be combined with post hoc interpretability tools to examine factor contributions (Castelvecchi, 2016). This robustness is particularly important for operational hazard mapping in reservoir regions, where reliable historical landslide inventories are often limited or incomplete. Furthermore, the inherent interpretability of tree-based models provides

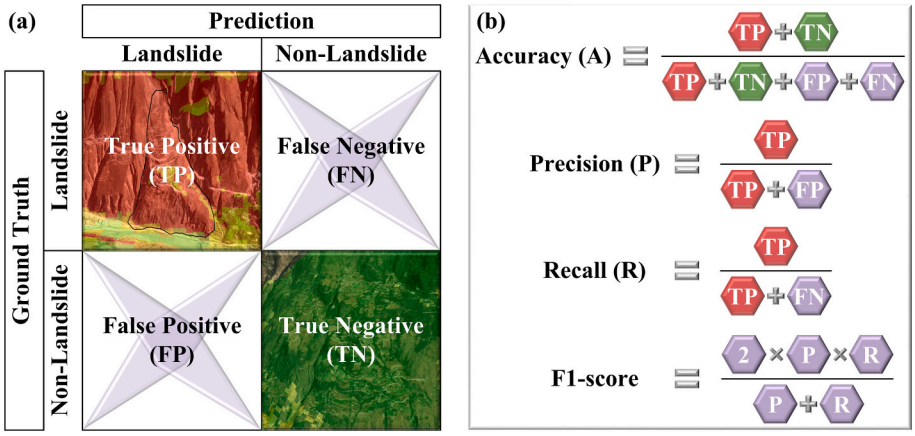

Fig. 7. (a) Confusion matrix. (b) Evaluation metrics.

clearer insight into feature interactions for decision support, enabling practitioners and decision-makers to validate risk assessments based on physical and geological principles before implementing mitigation strategies. The models were implemented in Python using scikit-learn; the workflow used a normalized dataset with a 7:3 train-test split (17,716 training vs. 7640 testing samples). Following hyperparameter optimization using Grid Search, each algorithm was trained across both negative sampling strategies, resulting in six comparative models (DT/RF/GBDT-InSAR and DT/RF/GBDT-IVM).

Model performance was evaluated using Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) (Ciampalini et al., 2016). Additional metrics, Accuracy, Precision, Recall, and F1-score, were also calculated (Fig. 7) (Ma et al., 2025).

3.5. State-based identification of PLAs

An integrated framework was established to identify PLAs by integrating static susceptibility with dynamic deformation monitoring. Unlike direct time-series forecasting, this framework represents hazard in a state-based manner at each observation stage by coupling inherent environmental predisposition with InSAR-derived kinematic signals. To maintain the non-linear grouping of geological data and avoid arbitrarily splitting high-risk clusters, the Jenks Natural Breaks optimization method was applied to discretize continuous model outputs. Furthermore, two fusion strategies were evaluated to delineate PLAs (Özcan et al., 2025).

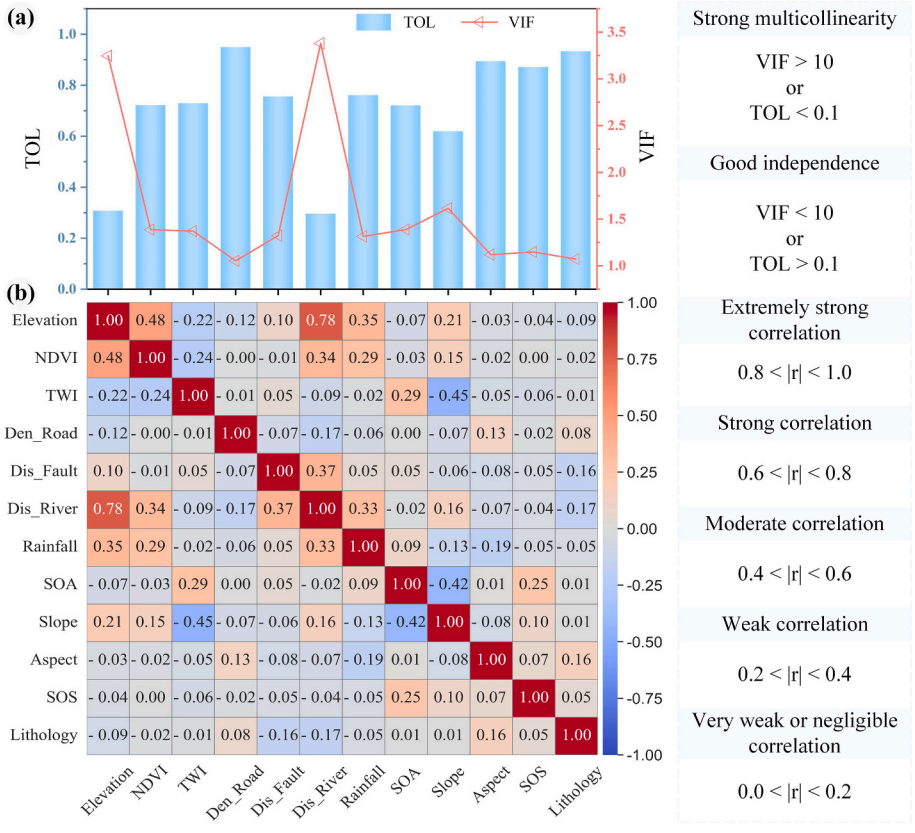

Fig. 8. Analysis of conditioning factors. (a) Collinearity diagnosis. (b) Pearson correlation.

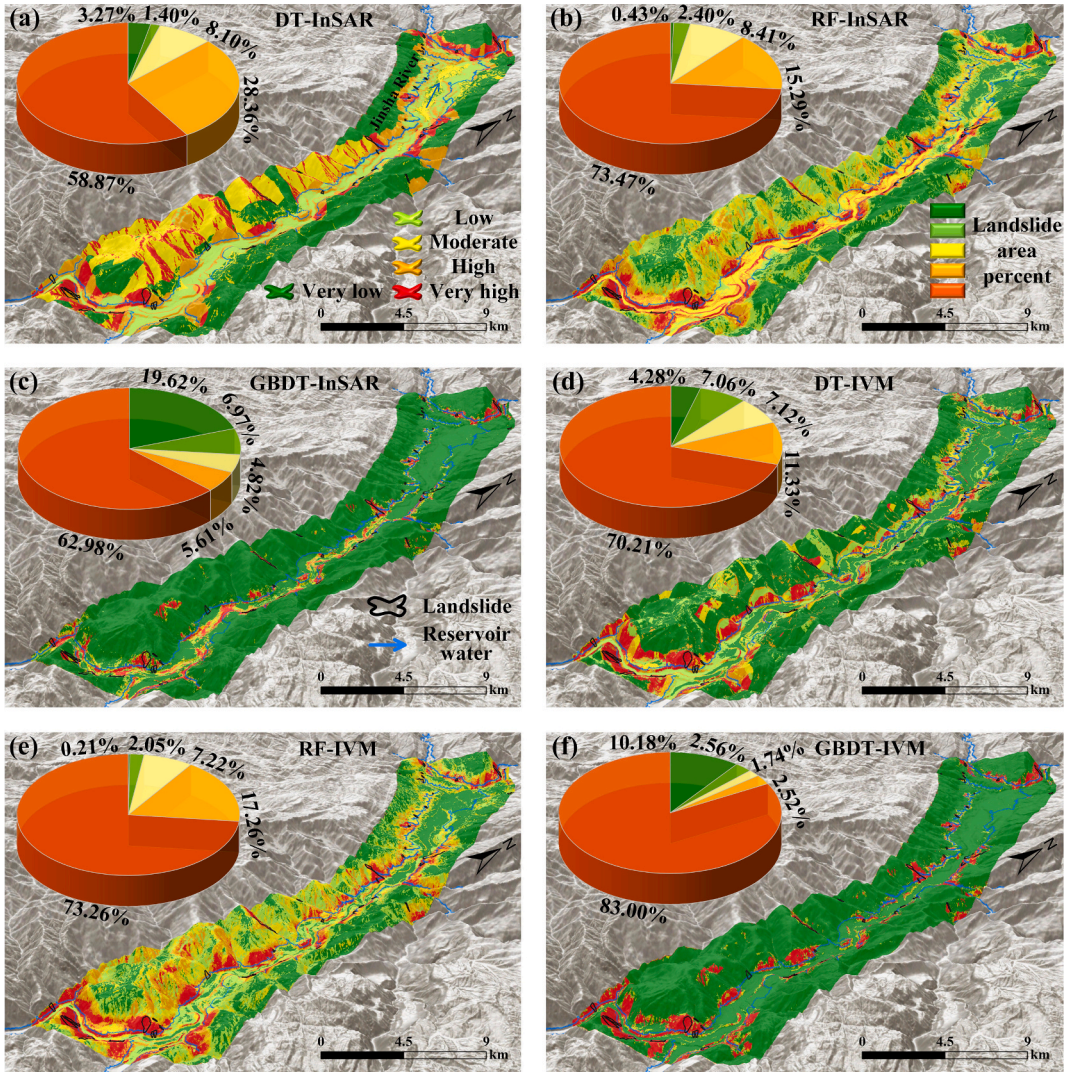

Fig. 9. Landslide susceptibility maps derived from different negative sampling strategies. (a-c) InSAR-based sampling. (d-f) IVM-based sampling. Pie charts show the proportion of inventoried landslide areas captured by respective susceptibility classes.

(1) Threshold Intersection Method: This strategy spatially overlays predefined deformation and susceptibility extremes. After masking decorrelated or low-quality pixels, we defined a stable LOS deformation threshold of ±10 mm/yr to account for Sentinel-1 measurement uncertainties and atmospheric noise in mountainous terrain (Yao et al., 2025). Pixels exhibiting active deformation coupled with "High" or "Very High" susceptibility (based on five Jenks classes) were identified as PLAs, followed by spatial clustering to reduce spatial noise and refine hazard boundaries.

(2) Contingency Matrix Strategy: Both InSAR measurements and susceptibility outputs were categorized into five Jenks classes (V1-V5 and S1-S5, respectively). Deformation tiers were delineated as V5 (−132 to −34 mm/yr), V4 (−34 to −16 mm/yr), V3 (−16 to −9 mm/yr), V2 (−9 to −3 mm/yr), and V1 (remaining stable points). Multi-geometry LOS deformation rates were directly coupled with the susceptibility index via a contingency matrix to maximize PLA detection. Subsequently, 2D decomposition of these LOS signals served as a diagnostic phase to characterize the specific kinematic mechanisms of the identified PLAs. The framework was evaluated by systematically comparing these two methodologies based on mapping accuracy and spatial coverage to ascertain the most effective strategy for identifying PLAs.

4. Results and analysis

4.1. Factor analysis and LSM with optimized negative sampling

Multicollinearity diagnostics for the 12 conditioning factors revealed acceptable levels of correlation. The maximum VIF was 3.38, and the minimum TOL was 0.30 (Fig. 8a), suggesting no significant multicollinearity. Pairwise Pearson correlation coefficients further supported this result, with a maximum correlation coefficient (r) of 0.78 (Fig. 8b). Consequently, all 12 factors, covering four domains: topography/geomorphology, hydrological environment, land cover, and lithological structure, were retained for subsequent susceptibility modeling. All factors were normalized prior to input into the ML model. This processing framework formed the basis for LSM.

The resultant susceptibility maps (Fig. 9) align with known geological patterns, with high-probability zones concentrating in areas governed by active destabilizing mechanisms rather than stochastic noise. (1) Hydrological-lithological coupling: The models effectively captured the instability along the Jinsha River, where reservoir water-level fluctuations degrade the shear strength of mechanically weak Quaternary

Table 2
Landslide susceptibility zoning results of optimized samples.

Model	Susceptibility level	Classification area/ km^2	Graded area percentage/%	Landslide area/ km^2	Landslide Surface density
DT-InSAR	Very low	241.38	38.14	0.3726	0.0015
	Low	92.77	14.66	0.1602	0.0017
	Moderate	96.56	15.26	0.9243	0.0096
	High	119.32	18.85	3.2355	0.0271
	Very high	82.80	13.09	6.7176	0.0811
RF-InSAR	Very low	219.38	34.67	0.0495	0.0002
	Low	145.05	22.92	0.2745	0.0019
	Moderate	117.92	18.64	0.9594	0.0081
	High	77.09	12.18	1.7442	0.0226
	Very high	73.36	11.59	8.3826	0.1143
GBDT-InSAR	Very low	531.52	83.99	2.2383	0.0042
	Low	23.87	3.77	0.7956	0.0333
	Moderate	16.43	2.60	0.5499	0.0335
	High	17.18	2.71	0.6399	0.0372
	Very high	43.82	6.93	7.1865	0.1640
DT-IVM	Very low	393.47	62.18	0.4887	0.0012
	Low	91.43	14.45	0.8055	0.0088
	Moderate	37.48	5.92	0.8118	0.0217
	High	40.69	6.43	1.2933	0.0318
	Very high	69.76	11.02	8.0109	0.1148
RF-IVM	Very low	219.76	34.73	0.0234	0.0001
	Low	158.47	25.04	0.2340	0.0015
	Moderate	102.21	16.15	0.8244	0.0081
	High	77.82	12.30	1.9692	0.0253
	Very high	74.57	11.78	8.3592	0.1121
GBDT-IVM	Very low	520.92	82.32	1.1619	0.0022
	Low	15.94	2.52	0.2925	0.0184
	Moderate	11.18	1.77	0.1989	0.0178
	High	13.38	2.11	0.2871	0.0215
	Very high	71.40	11.28	9.4698	0.1326

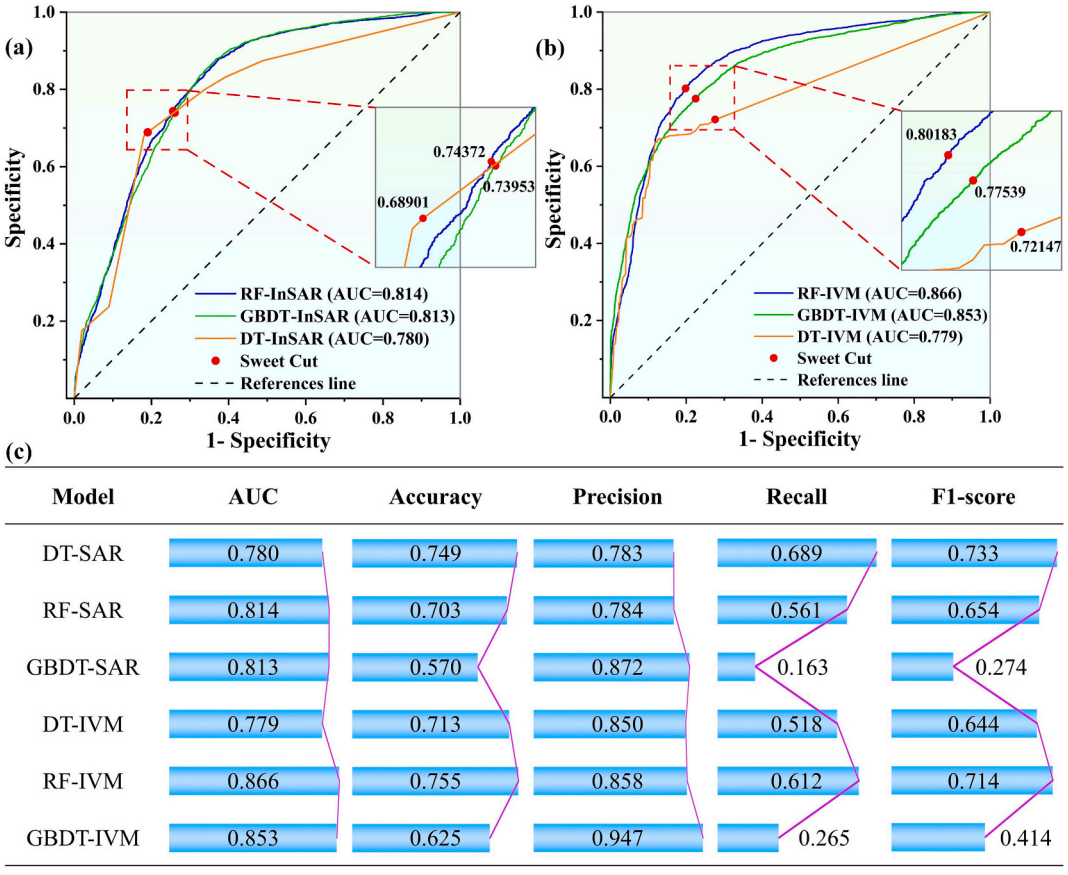

Fig. 10. ROC curves for models trained with (a) InSAR-based and (b) IVM-based negative sampling. (c) Comparison of performance metrics.

deposits and interbedded soft rocks. (2) Anthropogenic forcing: A noticeable spatial proximity between high-risk zones and road networks

suggests that slope toe excavation and stress redistribution associated with road construction may contribute to localized instability.

A quantitative comparison of the two sampling strategies revealed that the IVM-based approach significantly outperformed the InSAR-based sampling strategy, a trend particularly pronounced among ensemble classifiers (Table 2). While InSAR-constrained models yielded reasonable detection rates, they remained prone to elevated false-negative errors in complex terrain due to inherent geometric distortions like shadowing and layover. By incorporating statistical correlations to circumvent these observational gaps, the IVM strategy boosted the recall of unstable slopes. The RF-IVM model achieved the highest predictive performance, capturing 90.52% of the validated landslide area within the high and very high susceptibility classes.

Sampling strategy also had a strong influence on algorithmic bias. The GBDT algorithm, which showed a marked conservative bias (underprediction) under the InSAR-based protocol, exhibited an expansion of the mapped high-risk area from approximately 61 km^2 to 85 km^2 when IVM-selected samples were used, bringing the predictions into closer agreement with the observed hazard distribution. Overall, the RF-IVM configuration offered the most robust trade-off between sensitivity and specificity, delineating a refined high-risk zone of 152.39 km^2. This underscores a fundamental insight: optimizing the quality of negative samples via statistical methods is as important as algorithm selection, as it provides a more reliable basis for the model to capture meaningful and spatially consistent environmental drivers of landslide susceptibility.

4.2. Model performance evaluation

The ROC analysis indicates that ensemble learning algorithms (RF, GBDT) consistently outperformed the single DT model, reflecting the capacity of bagging and boosting mechanisms to mitigate overfitting (Fig. 10). Among all configurations, RF-IVM yielded the highest AUC (0.866), indicating the strongest overall discriminative ability and robust generalization. Crucially, the optimization of negative sampling strategies proved to be a decisive factor for advanced classifiers: the transition from InSAR-based to IVM-based sampling led to measurable performance gains ($\Delta AUC_{RF} = +0.052$; $\Delta AUC_{GBDT} = +0.040$). This improvement likely reflects the IVM's ability to filter noisy stable pixels, thereby removing ambiguous samples that appear deformation-stable yet environmentally predisposed to failure, and ultimately sharpening the decision boundary between landslide and non-landslide classes.

While global AUC metrics suggest high performance, the confusion matrix-derived metrics reveal distinct operational characteristics and trade-offs among the classifiers (Fig. 10c). The GBDT model, despite attaining the highest precision (0.947 under IVM), exhibited highly conservative classification behavior characterized by very low recall (0.265). This imbalance implies that GBDT minimizes false positives at the expense of missing a majority of actual landslides, rendering it less suitable for regional hazard screening where omission errors are particularly undesirable.

In contrast, the RF model delivered the most balanced performance between precision and recall. Achieving the highest F1-score (0.714) under the IVM strategy, RF maintained moderate recall (0.612) without compromising precision, supporting its suitability for regional-scale zonation where both detection completeness and reliability are important. Meanwhile, the single DT showed negligible sensitivity to sampling optimization and generally lower AUCs, underscoring its limited ability to represent the complex, non-linear geological interactions captured by ensemble methods.

4.3. Potential landslide area identification by integrating InSAR deformation rates with susceptibility zoning

Using the combined benchmark of the model development inventory and the independent validation set, the two fusion strategies showed clear performance differences (Tables 3 and 4). Constrained by fixed binary thresholds for both velocity and susceptibility, the threshold intersection method was overly conservative, yielding a recall of only 59.40% (44/74) and a spatial overlap of 15.44% with the reference landslide inventory. Conversely, the contingency matrix strategy showed substantially better performance, correctly identifying 90.50% (67/74) of the inventory events while increasing the overlap area with mapped landslides by more than fourfold, with a spatial overlap of 63.95%. By conceptualizing deformation and susceptibility as coupled gradients rather than independent binary filters, the matrix-based approach yielded coherent hazard zones that align closely with the morphological boundaries of actual slope failures, such as the Huangping and Wulipo landslides (Fig. 11).

Table 3
Numbers of known landslides identified by different methods in the model development inventory and the independent validation set.

	Model development inventory	Independent validation set
Landslide susceptibility	59	15
InSAR deformation rate	20	5
Threshold model	33	11
Contingency Matrix	52	15

Table 4
Comparison of landslide areas identified by the two PLA delineation strategies.

	Potential landslide area /km^2	Overlap area with mapped landslides	Overlap area within the independent validation set
Threshold model	7.91	1.76	0.63
Contingency Matrix	76.39	7.29	4.68

To better understand the mechanisms underlying this performance, we analyzed three representative coupling scenarios (Fig. 12) that illustrate how the matrix approach overcomes the limitations of single-source monitoring:

Scenario I: Sensitivity to slow-moving or dormant hazards (high susceptibility + low deformation). This scenario demonstrates the effectiveness of the method in areas affected by low coherence or decorrelation gaps. The slope in Fig. 12a exhibits high geological predisposition (lithology/slope) but lacks significant InSAR-detectable deformation, likely due to temporal intermittency or geometric masking. The contingency matrix correctly classified it as a PLA by allowing the "Very High" susceptibility score to compensate for the weak deformation signal. This suggests that the method can capture dormant or creeping hazards that elude purely deformation-based criteria.

Scenario II: Suppression of anthropogenic noise (low susceptibility + high deformation). Fig. 12b depicts a zone of intensive construction and excavation, showing high surface velocity (−39 mm/yr). A single-source InSAR approach could misclassify this area as a landslide. Crucially, the fusion model rejected this area (classifying it as non-PLA) because the environmental background (flat terrain, stable lithology) yielded a "Low" susceptibility score. This suggests that the model can reduce false positives associated with anthropogenic surface settlement.

Scenario III: Confirmation of active instability (high susceptibility + high deformation). Fig. 12c represents the ideal convergence, where reservoir-induced toe erosion drives rapid deformation (−40 mm/yr) on a geologically fragile slope. The matrix integration reinforced the hazard signal, assigning the highest probability level.

These scenarios indicate that the contingency matrix does more than simply overlay datasets; it functions as a compensatory decision framework. It recovers missed detections in geologically hazardous but

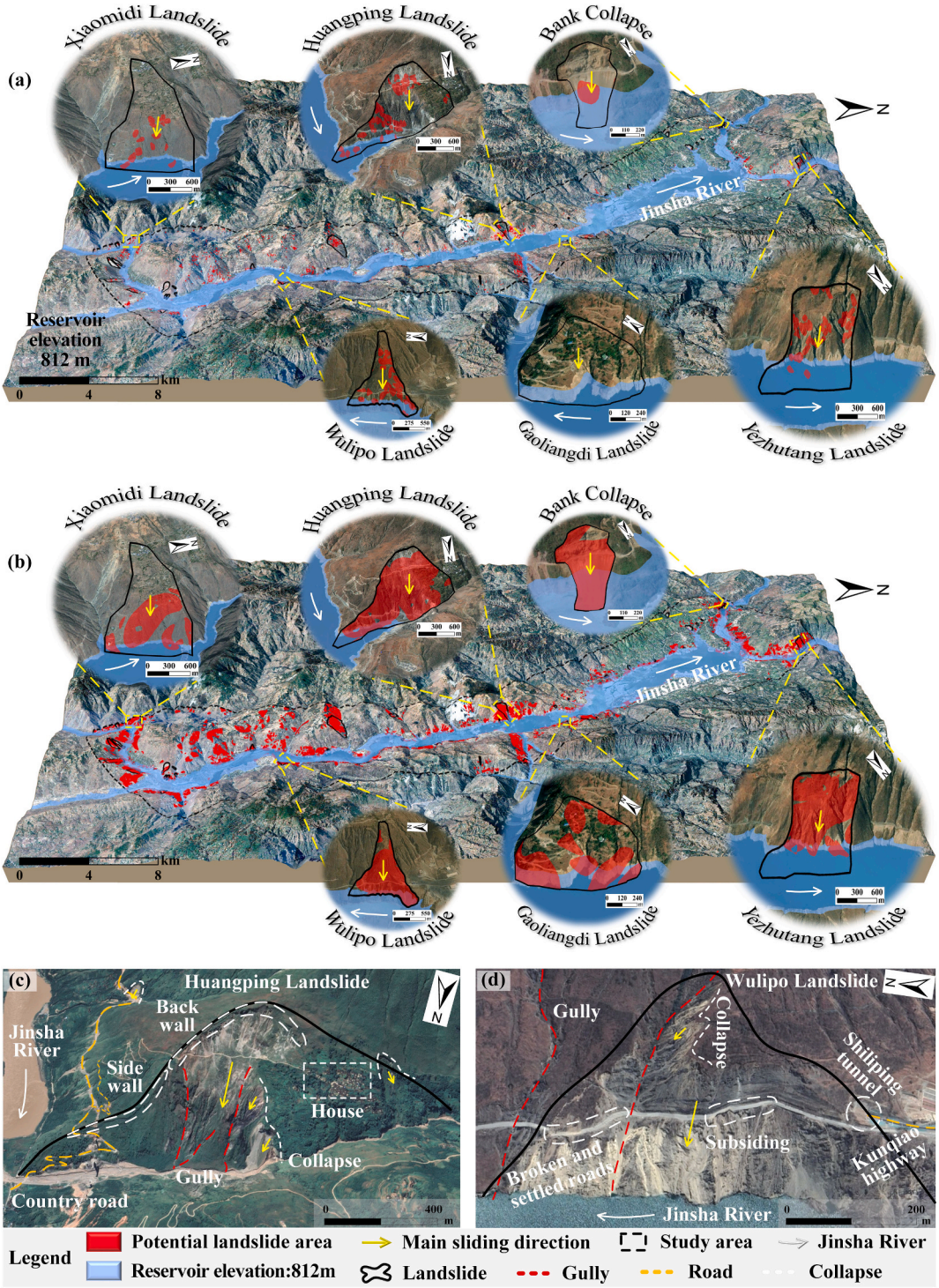

Fig. 11. PLA identification using different fusion strategies. (a) Threshold intersection and (b) Contingency matrix. Detailed views of the (c) Huangping and (d) Wulipo landslides.

InSAR-insensitive areas (Scenario I) while suppressing false alarms in anthropogenically disturbed but geologically stable zones (Scenario II).

4.4. Kinematic evaluation of PLAs based on 2D decomposition

While the contingency matrix in Section 3.5 efficiently identifies potential hazards using LOS data, 1D measurements cannot fully resolve the true movement direction of complex slopes. Therefore, this section applies 2D decomposition to analyze the vertical and horizontal components of the previously identified PLAs, providing a kinematic basis for evaluating whether the model-predicted high-susceptibility zones exhibit failure modes, such as subsidence-driven bank collapse, that are characteristic of reservoir-induced hydro-mechanical forcing. This step extends the assessment beyond purely probabilistic zonation by adding post hoc kinematic verification to the identified PLAs (Fig. 13). The vertical velocity component (V_U) showed stronger localized subsidence, reaching −93 mm/yr, whereas the horizontal component (V_E) ranged from −55 to +102 mm/yr, suggesting that vertical

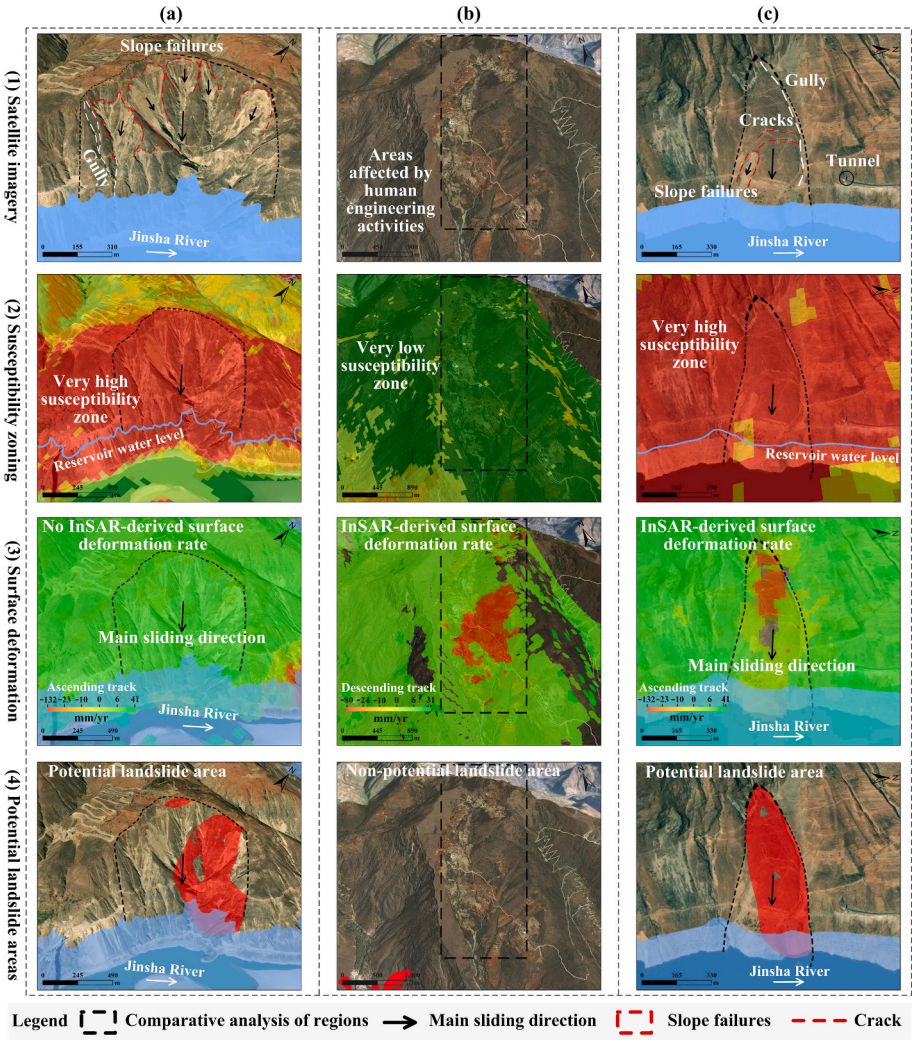

Fig. 12. Identification of slopes under different coupling scenarios (Rows 1–4: Optical imagery, Susceptibility, LOS deformation, PLAs). Columns correspond to: (a) High susceptibility but no surface deformation. (b) Low susceptibility with significant deformation. (c) High susceptibility with pronounced deformation.

deformation plays an important role in slope instability within the reservoir region. Crucially, the contingency matrix effectively filtered non-geological noise. Despite localized LOS velocities exceeding −40 mm/yr near infrastructure developments (e.g., excavation at Point 12, road construction at Point 17), these areas were correctly classified as low hazard because they lacked predisposing geological conditions, effectively preventing the misclassification of anthropogenic disturbances, as shown in Fig. 14.

The decomposition of 2D displacement vectors provided diagnostic insight into failure mechanisms and revealed clear links to reservoir hydrodynamics (Fig. 15). For the instability at Point 21, the kinematic signature is characterized by dominant vertical subsidence (Max V_U = −55.14 mm/yr) over horizontal motion. This vertical dominance is consistent with a bank-collapse-dominated or subsidence-dominated deformation mode, possibly associated with toe erosion related to fluvial scour along the Jinsha River (Fig. 15a). In contrast, the Wulipo landslide (Point 24) and Point 28 exhibit complex multidirectional kinematics characteristic of translational failure. At Wulipo, pronounced westward horizontal displacement (V_E = −51.85 mm/yr), together with vertical settlement (V_U = −38.86 mm/yr), indicates a rotational-translational movement pattern, with an overall displacement vector directed northwestward toward the free slope facing the river.

From a spatial perspective, the distribution of these deformation hotspots is also informative: the maximum deformation zones are concentrated within a distinct elevation band around 835 m. This band is adjacent to the reservoir water-level fluctuation zone, suggesting that hydro-mechanical forcing, particularly periodic saturation and drawdown cycles, is a primary driver of the observed instability. This correspondence further supports the interpretation that the predicted PLAs effectively capture active, water-driven kinematic zones.

Comprehensive field investigations and UAV surveys support the physical reliability of the InSAR-derived signals (Fig. 16), showing that high-velocity zones are spatially associated with distinct active geomorphic indicators.

At the Xiaomidi Landslide (Point 1), the extreme LOS velocity (Max V_{LOS} = −131 mm/yr) corresponds directly to a slope destabilized by anthropogenic disturbance and loose Quaternary deposits, as indicated by exposed bedrock and well-developed erosional gullies. Similarly, in the vicinity of the Wanliang Tunnel (Point 19), InSAR-derived rates of −40 mm/yr are consistent with the structural distress observed in the field, where prominent tensile cracks, some reaching 18 cm in width, extend across the slope and may compromise the integrity of the overburden. At the Wangjiashan Landslide (Point 23), active mitigation measures, including the installation of rockfall protection nets and monitoring instruments along stepped cuts, further support the satellite-detected displacement signal of −53 mm/yr. Finally, the kinematic signature of the Wulipo Landslide (Point 24) reflects ongoing material transport, characterized by overall subsidence (Max V_{LOS} = −60 mm/

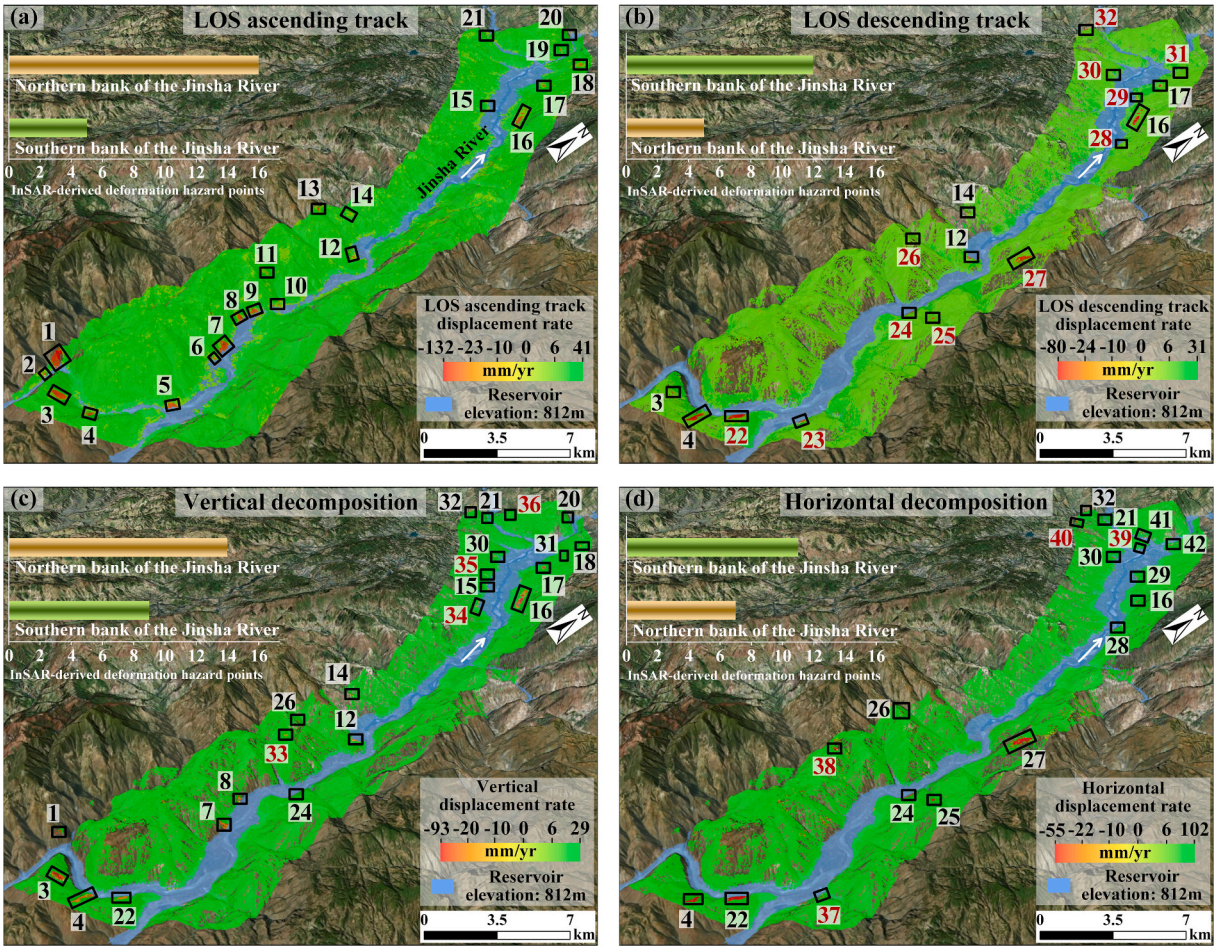

Fig. 13. Kinematic observations. (a–b) SBAS-InSAR LOS deformation rates. Decomposed (c) horizontal and (d) vertical deformation rates. Red values indicate localized velocity increases revealed by 2D decomposition compared to 1D LOS data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

yr), which is consistent with the accumulation of weathered colluvium in the middle-lower slope sections.

5. Discussion

Unlike conventional approaches that treat InSAR deformation merely as an auxiliary weighting factor or posterior adjustment, the proposed framework couples static susceptibility (a predisposing condition) with dynamic deformation (a state indicator) through a physically constrained contingency matrix. This state-based formulation identifies high-risk zones where active deformation coincides with geological predisposition. In addition, SHAP integration provides a transparent and physically interpretable link between hydro-mechanical drivers and hazard zoning.

5.1. Model interpretability and geophysical insights from SHAP analysis

The SHAP framework quantifies the physical drivers of instability, thereby bridging the gap between algorithmic predictions and a mechanistic understanding of how topography, lithology, and hydrology interact to influence landslide susceptibility (Yao et al., 2024). Unlike the intrinsic feature importance of RF, which prioritizes static topographic conditions (Slope), SHAP identifies Dis_River as the most influential predictor in the model (Fig. 17). This suggests that, in this study area, reservoir-related hydraulic boundary effects may be more influential than some local topographic proxies.

The present SHAP analysis evaluates the contribution of conditioning factors at a fixed mapping stage rather than providing a time-resolved interpretation of slope evolution. Because reservoir hydrodynamics vary over time, a state-dependent interpretation would require temporally segmented datasets (e.g., separating the impoundment and drawdown phases) and corresponding model recalibration for each period. In this study, the dynamic behavior of slopes is instead captured through InSAR-derived deformation and its integration with static susceptibility via the contingency matrix.

Physically, high SHAP values for river proximity (< 500 m) serve as a proxy for the reservoir hydro-fluctuation belt (Fig. 18a). Periodic wetting-drying cycles within this zone trigger two destabilizing processes: mechanical toe erosion, which removes kinematic support, and rapid drawdown, which induces transient hydraulic gradients and reduces the shear strength of saturated soils (Tang et al., 2019). Field validation and monitoring data support this mechanism, with active tension cracks and toe collapses concentrated almost exclusively within this SHAP-defined high-risk band (Fig. 19).

The secondary importance of Lithology and Dis_Fault highlights the role of material degradation. The negative SHAP trend with Dis_Fault reflects tectonic damage zones, where high fracture density increases secondary permeability. This structural fragmentation facilitates rapid groundwater infiltration into mechanically weak sandstone-mudstone interbeds (Fig. 19a-c), thereby accelerating slaking and softening, a process consistent with the hydro-mechanical coupling mechanism reported in similar active tectonic belts (Huang et al., 2012). Concurrently, low NDVI values (Fig. 18d), indicating sparse vegetation and reduced root cohesion, show a positive association with failure probability,

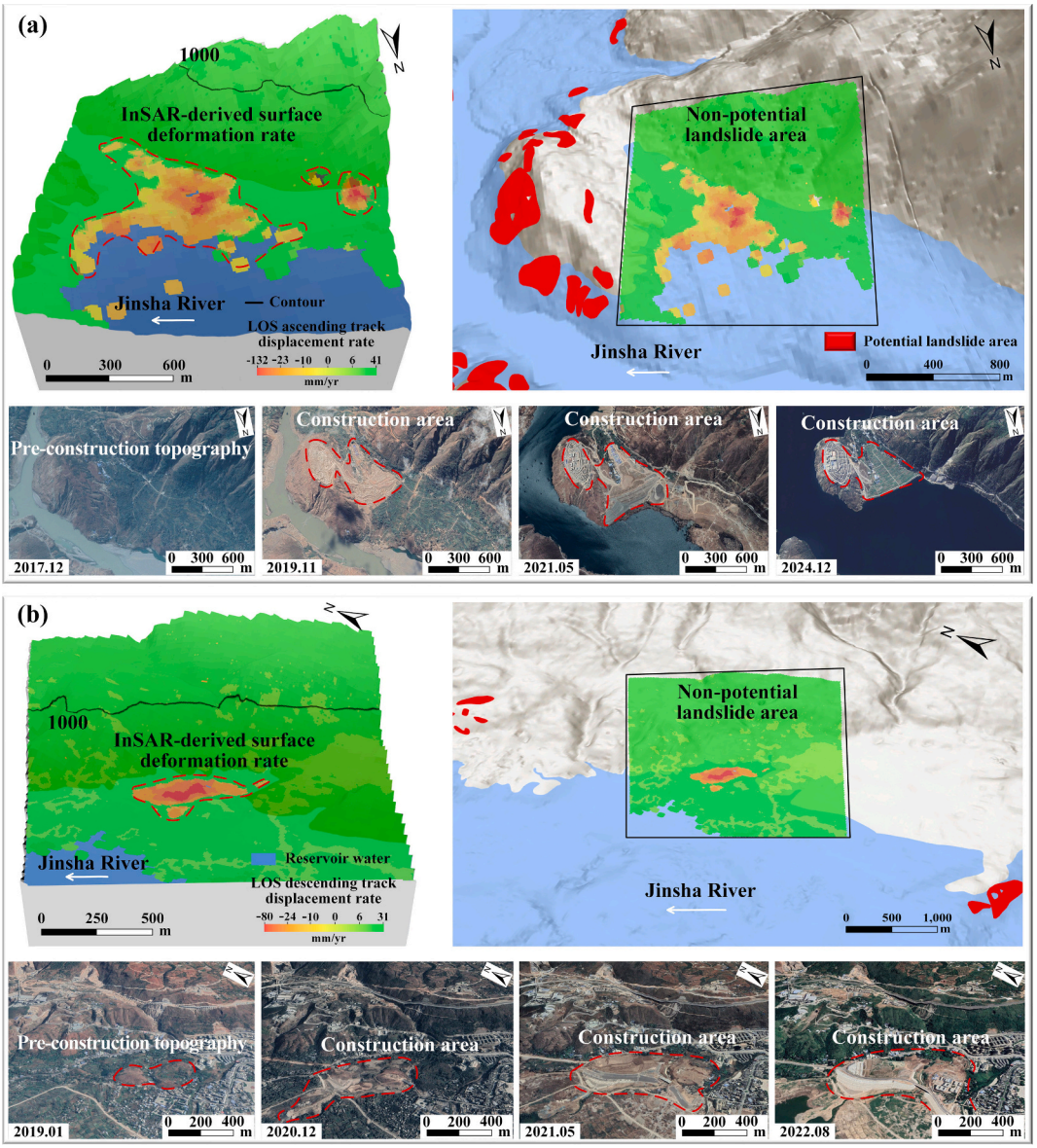

Fig. 14. InSAR-derived deformation rates and corresponding multi-temporal satellite imagery at representative locations. (a) Excavation at Point 12 and (b) road construction at Point 17.

consistent with previous findings in subtropical mountainous environments (Zhang et al., 2024a).

The bivariate SHAP dependence plots (Fig. 18e-i) further reveal nonlinear coupling effects that cannot be captured by single-factor analysis and are consistent with geomechanical principles. A synergistic amplification of risk is observed for slopes at lower elevations (< 800 m) close to the river. This combination defines an active mobilization zone, where the reservoir water-level (approx. 825 m) directly interacts with the slope toe. Soft rocks located near faults exhibit disproportionately high SHAP values. This pattern suggests that tectonic fracturing may facilitate weathering, increasing the specific surface area available for water-rock interaction and accelerating the degradation of rock mass strength parameters (Li and Moon, 2021).

Interestingly, factors such as Profile_c and TWI, which are generally considered important for landslide stability, showed relatively weak SHAP contributions in this study. This attenuation is likely attributable to the dominant influence of the reservoir boundary condition. Under this hydrological regime, the dynamic groundwater table is controlled more by lateral seepage from the reservoir (represented by Dis_River) than by topographic convergence of rainfall (represented by TWI). This suggests a fundamental hierarchy: in reservoir-induced landslide modeling, boundary hydraulic conditions may outweigh local topographic controls (Yin et al., 2016).

From a digital twin perspective, the SHAP-identified sensitive features provide a direct link between data-driven interpretation and operational monitoring. The identified threshold of 500 m from the river (Fig. 18a) serves as a spatial boundary for defining the active hydro-mechanical coupling zone. This boundary can guide the optimized deployment of in-situ sensors (e.g., piezometers and GNSS receivers) and support mesh refinement in hydrological simulation models (Qi et al., 2010). The strong coupling between Dis_Fault and Lithology suggests that digital twin models should prioritize dynamic parameter updating (e.g., strength degradation curves) for soft rock strata within the fluctuation zone, rather than applying uniform reduction factors (Wang et al., 2021). By incorporating these SHAP-identified sensitive features into the physics-based layer of the digital twin, the system can evolve from static susceptibility mapping toward dynamic, process-based early warning, with particular emphasis on the pre-failure window associated with rapid drawdown cycles (Al-Najjar et al., 2025).

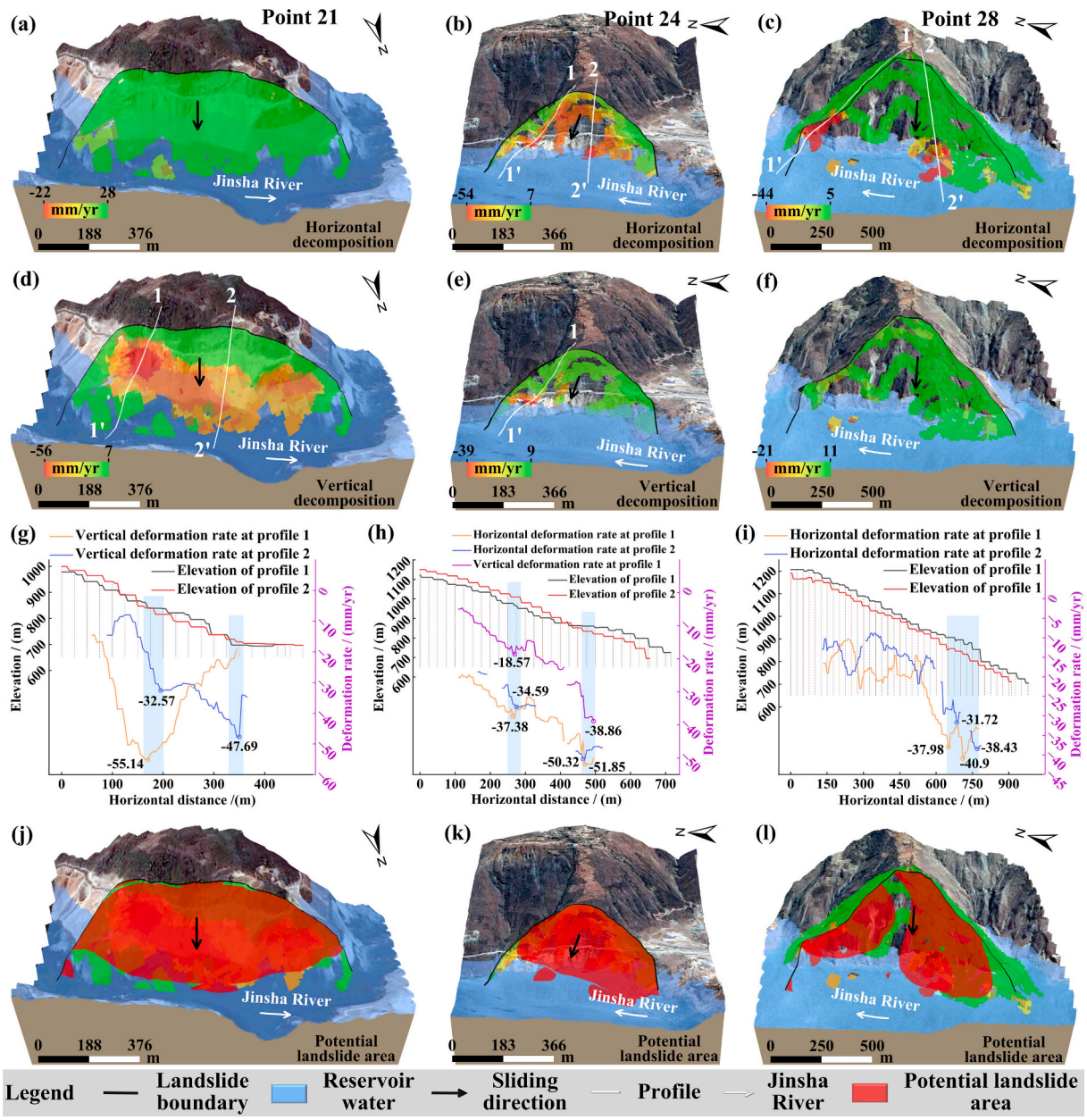

Fig. 15. 2D kinematic characteristics and PLA identification for representative slopes (Points 21, 24, 28). (a-c) Horizontal and (d-f) vertical deformation rates. (g-i) Elevation profiles and corresponding deformation rates along cross-sections 1–1′ and 2–2′ at Points 21, 24, and 28, respectively, illustrating the relationship between topography and deformation patterns. (j-l) Corresponding PLA delineations.

5.2. Model robustness and comparative performance

Comparative experiments reveal that model performance is closely linked to the negative sampling strategy, with the DT, RF, and GBDT algorithms exhibiting distinct trade-offs. The single DT model showed relatively strong initial performance under InSAR-based sampling because of the clear separability between active and stable slopes, but it failed to generalize under the more complex IVM-based sampling strategy, reflecting its limited capacity to represent nonlinear relationships. By contrast, the GBDT model displayed a highly conservative tendency: despite achieving high precision (0.872–0.947), its very low recall (0.163–0.265) suggests overfitting to a narrow subset of clearly expressed landslides, causing it to miss many subtle deformation patterns (Fig. 20a-d). By comparison, the RF-IVM model achieved the most balanced performance (AUC = 0.866; F1 = 0.714), effectively combining the variance-reduction advantage of ensemble learning with refined negative sampling. By removing ambiguous training data, the IVM filter reduced false positives by 3.1% and false negatives by 2.5%, thereby sharpening the decision boundary.

Under the IVM-based negative sampling strategy, the GBDT model exhibits high precision but relatively low recall compared with RF. This behavior can be attributed to the boosting-based learning mechanism of GBDT, which tends to produce conservative decision boundaries when trained on well-separated negative samples. The IVM strategy selects negative samples with clear feature-space separability, which helps reduce false positives but also increases the likelihood of missing marginal or transitional landslide cases. In contrast, the bagging-based RF model is more tolerant of sample variability and boundary uncertainty, resulting in a more balanced trade-off between precision and recall. These findings highlight an important implication: ensuring the reliability of negative samples is as pivotal as feature engineering for robust and physically meaningful susceptibility mapping (Daviran et al., 2023).

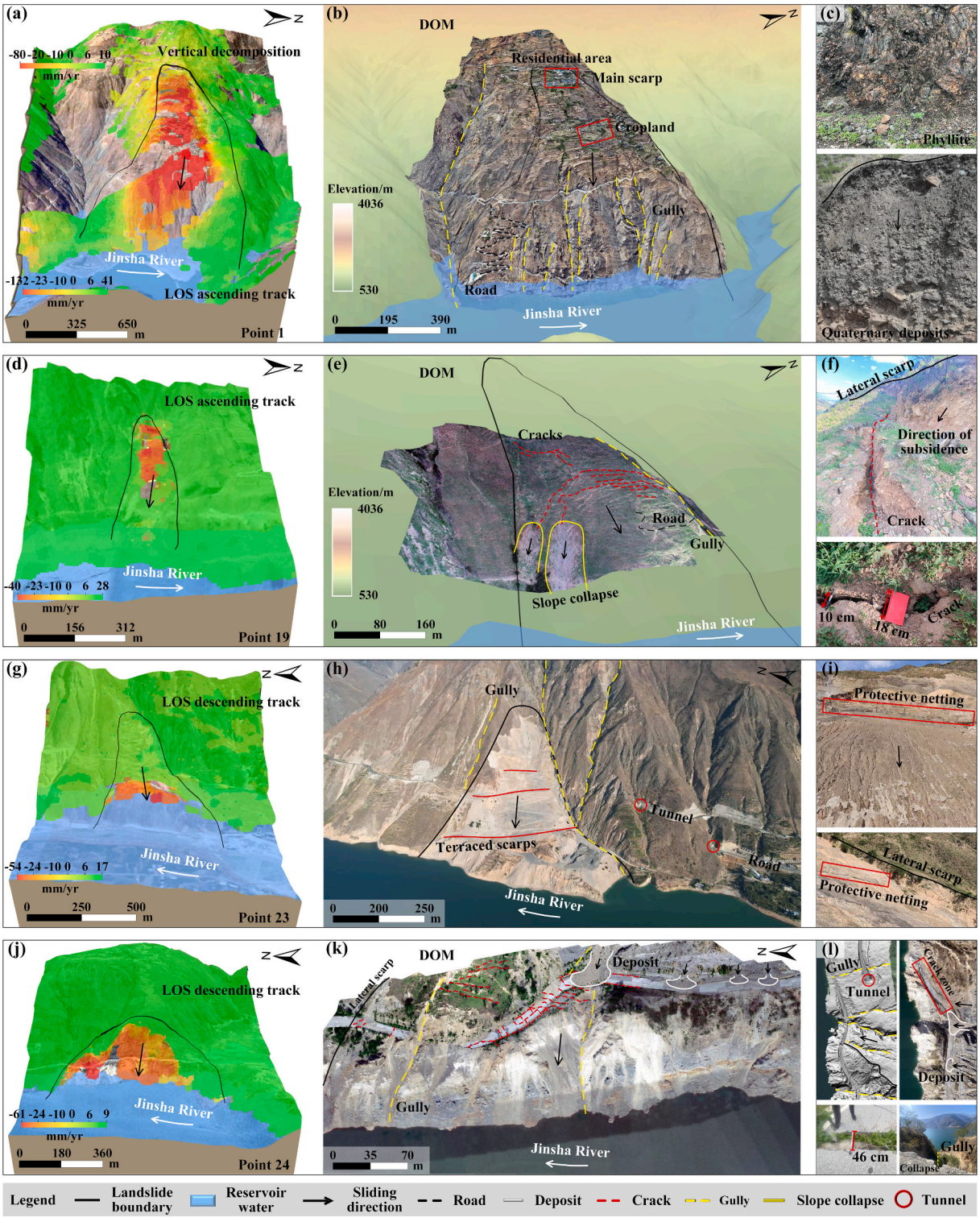

Fig. 16. Field validation and remote sensing evidence for identified PLAs. (a-c) Xiaomidi landslide (Point 1): LOS velocity, UAV orthophoto, and field survey. (d-f) Wanliang Tunnel landslide (Point 19): LOS velocity, UAV view of surface cracks, and field photo of scarp. (g-i) Wangjiashan landslide (Point 23): LOS velocity, UAV panorama, and slope cutting. (j-l) Wulipo landslide (Point 24): LOS velocity, UAV orthophoto, and LiDAR-derived roadbed subsidence.

However, although the IVM- and InSAR-based strategies improve negative sample purity, they also introduce a preset prior toward clearly stable geomorphic settings. This may underrepresent the background variability of the natural landscape and potentially lead to boundary overfitting. Future studies could adopt a hybrid sampling strategy that combines optimized samples with purely random background samples to balance local sample purity with broader environmental variability. A systematic comparison with such hybrid strategies was not conducted in the present study and warrants dedicated evaluation in future work.

From a computational perspective, the proposed contingency matrix approach introduces little additional computational complexity in practice compared with the conventional threshold-based model, as both rely on the same susceptibility outputs and InSAR-derived deformation products. The improved zoning performance and interpretability

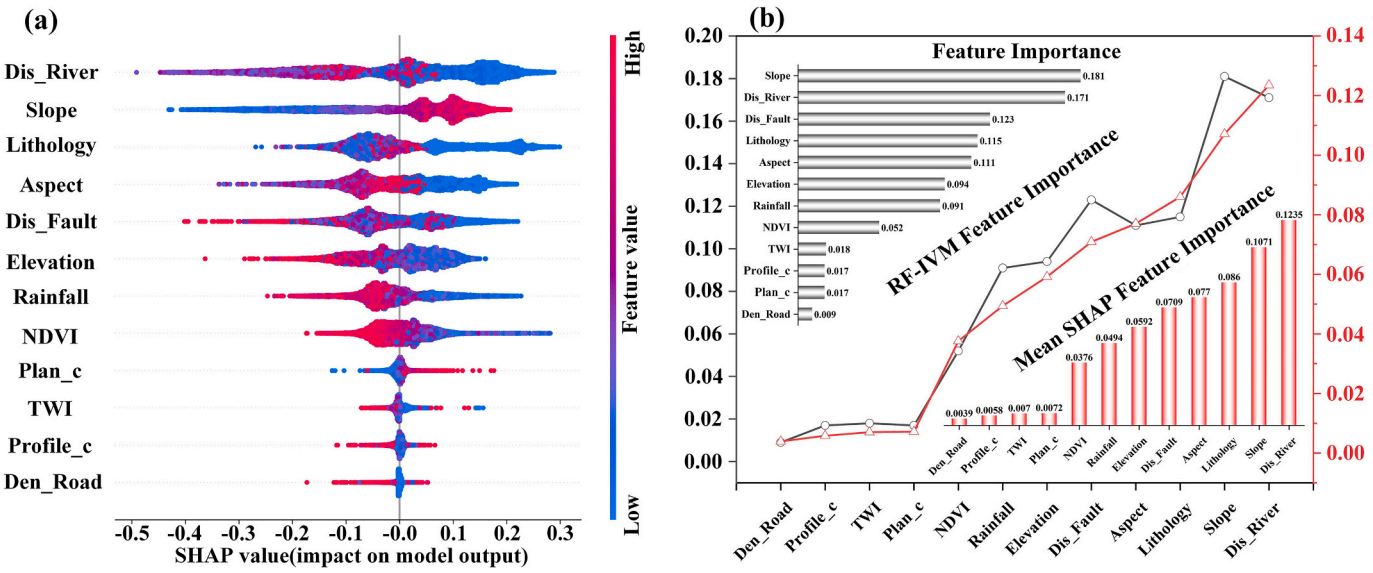

Fig. 17. SHAP interpretability analysis: (a) Summary plot and (b) feature importance ranking of conditioning factors.

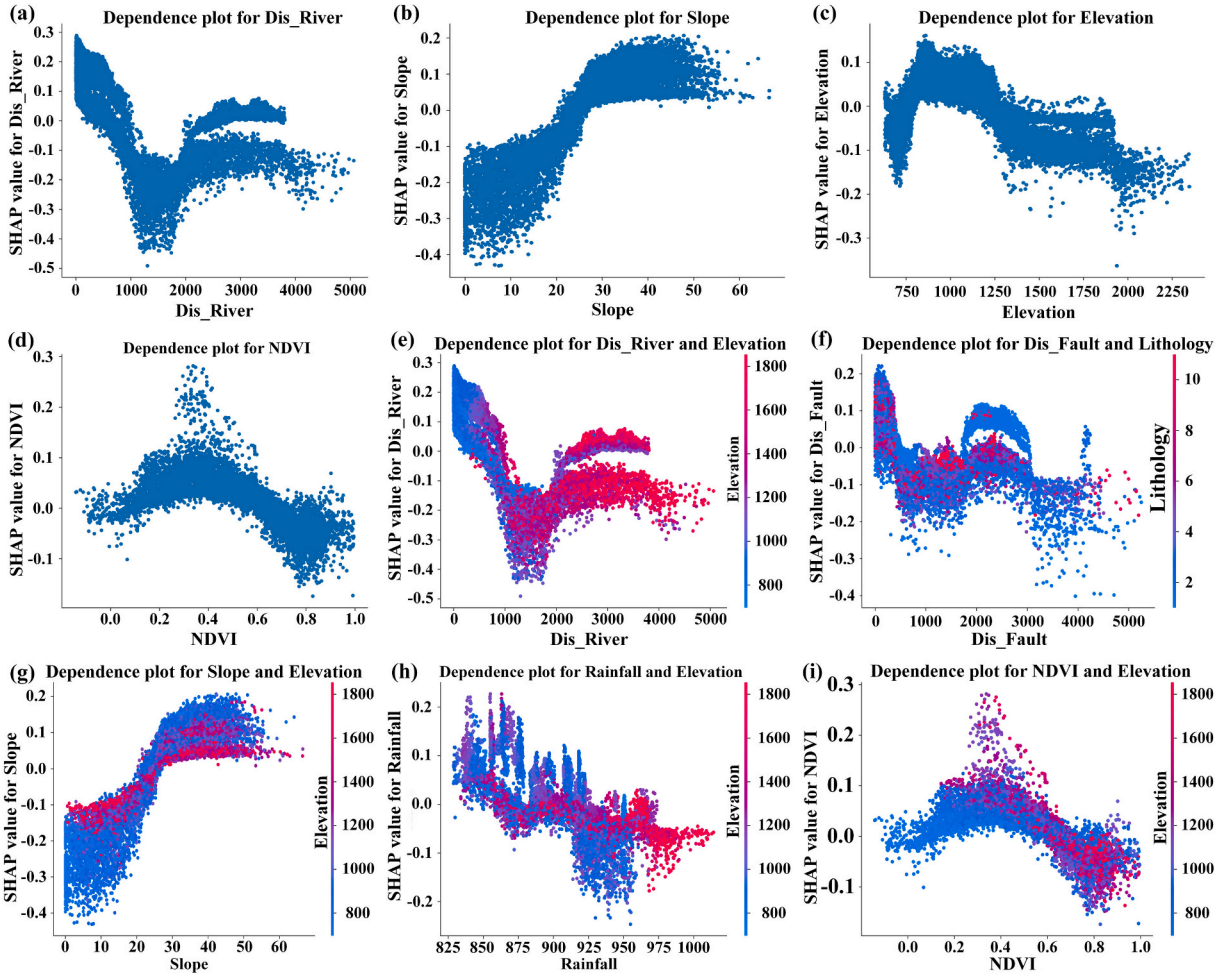

Fig. 18. SHAP dependence plots for key conditioning factors. (a-d) Main effects for distance to river, slope, elevation, and NDVI. (e-h) Bivariate interaction effects between elevation and distance to river, slope, rainfall, and NDVI. Color gradients represent the value range of the interacting variable.

are achieved without increasing computational burden, supporting the practical application of the contingency matrix approach for large-scale hazard screening.

5.3. Parameter robustness and boundary constraints

A qualitative comparison suggests that the adopted classification

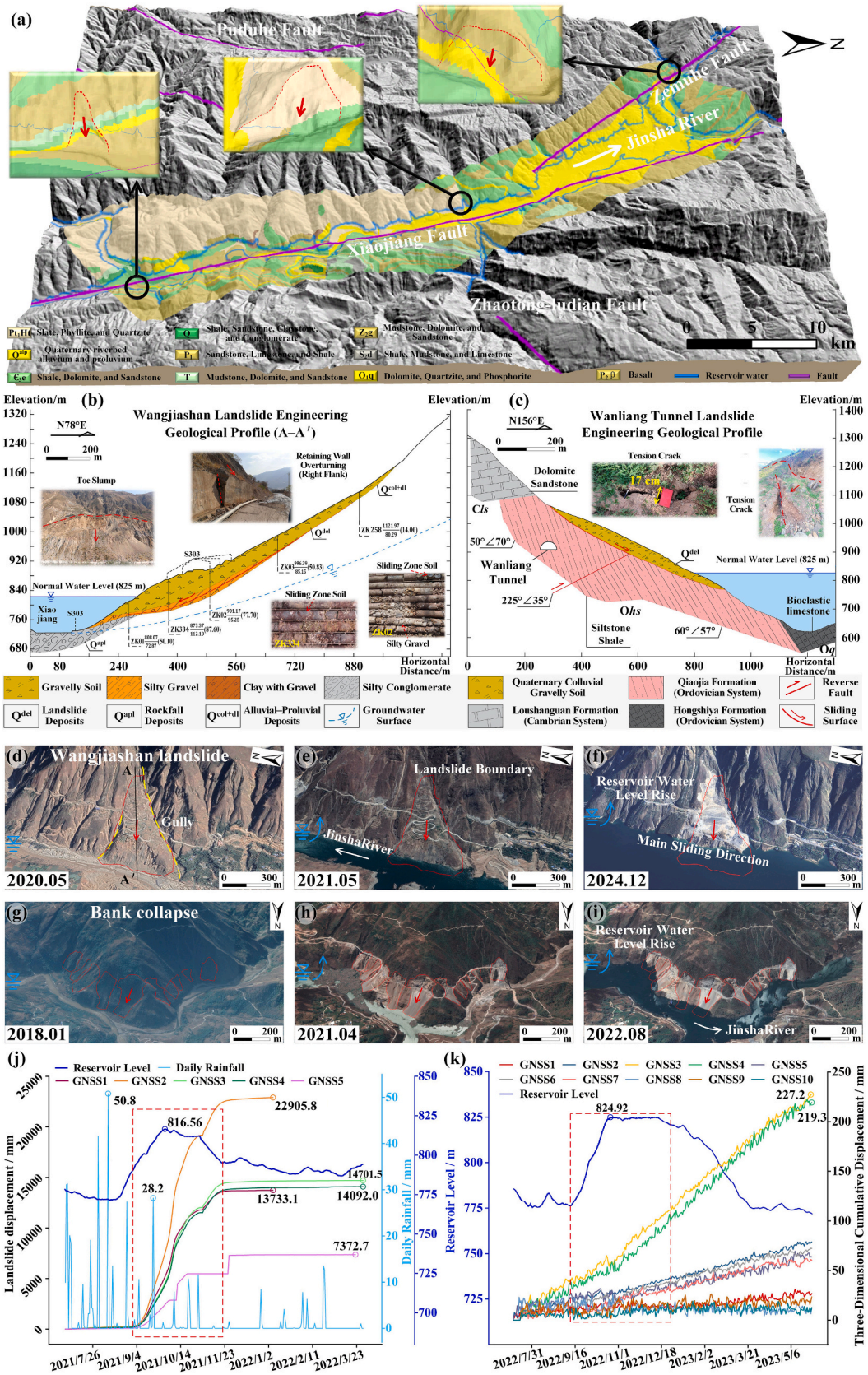

Fig. 19. Geological and observational validation of SHAP interpretations. (a) Regional lithology. (b-c) Geological profiles of the Wangjiashan and Wanliang Tunnel landslides. (d-f) Pre- and post-impoundment imagery of Wangjiashan. (g-i) Evolution of bank collapses. Time series of GNSS displacement, rainfall, and water-level for (j) Wangjiashan and (k) Xiaomidi landslides.

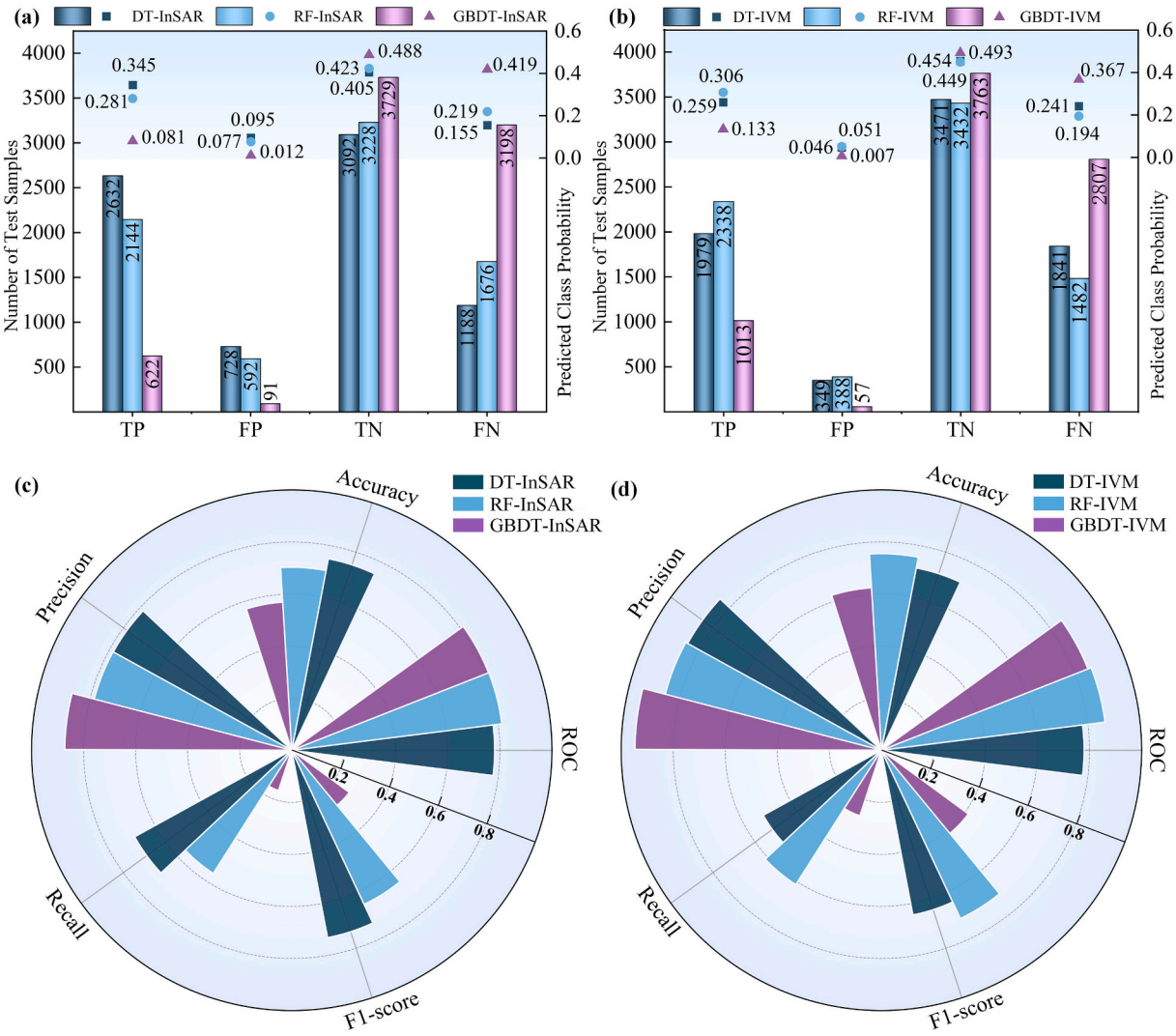

Fig. 20. Comparison of LSM performance. Predicted sample distributions under (a-b) InSAR and IVM sampling strategies. Evaluation metrics for models trained with (c) InSAR and (d) IVM sampling.

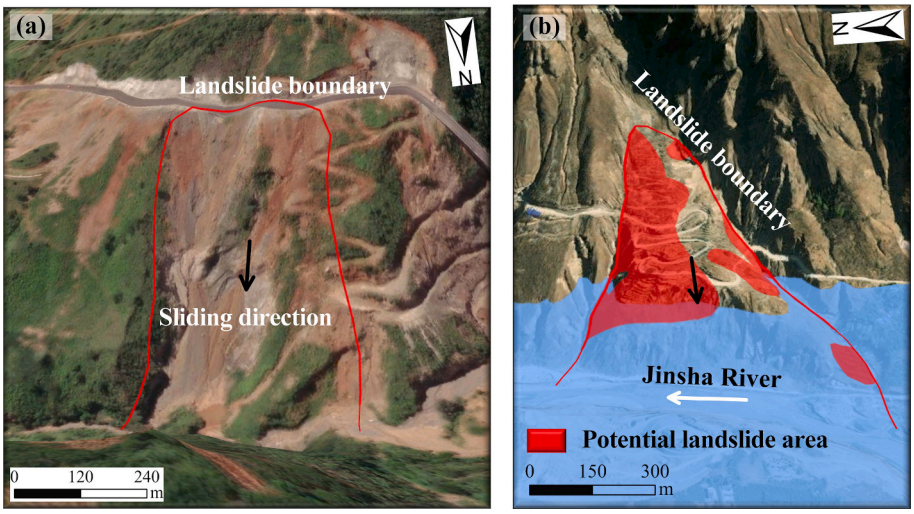

Fig. 21. Boundary cases illustrating PLA identification limitations: (a) Missed detection of a known landslide due to small scale or inconspicuous deformation. (b) Partial coverage of a landslide, reflecting spatial uncertainty and scale effects.

parameters are reasonably stable for the present dataset. Compared with linear classification schemes, which tend to over-fragment hazard zones

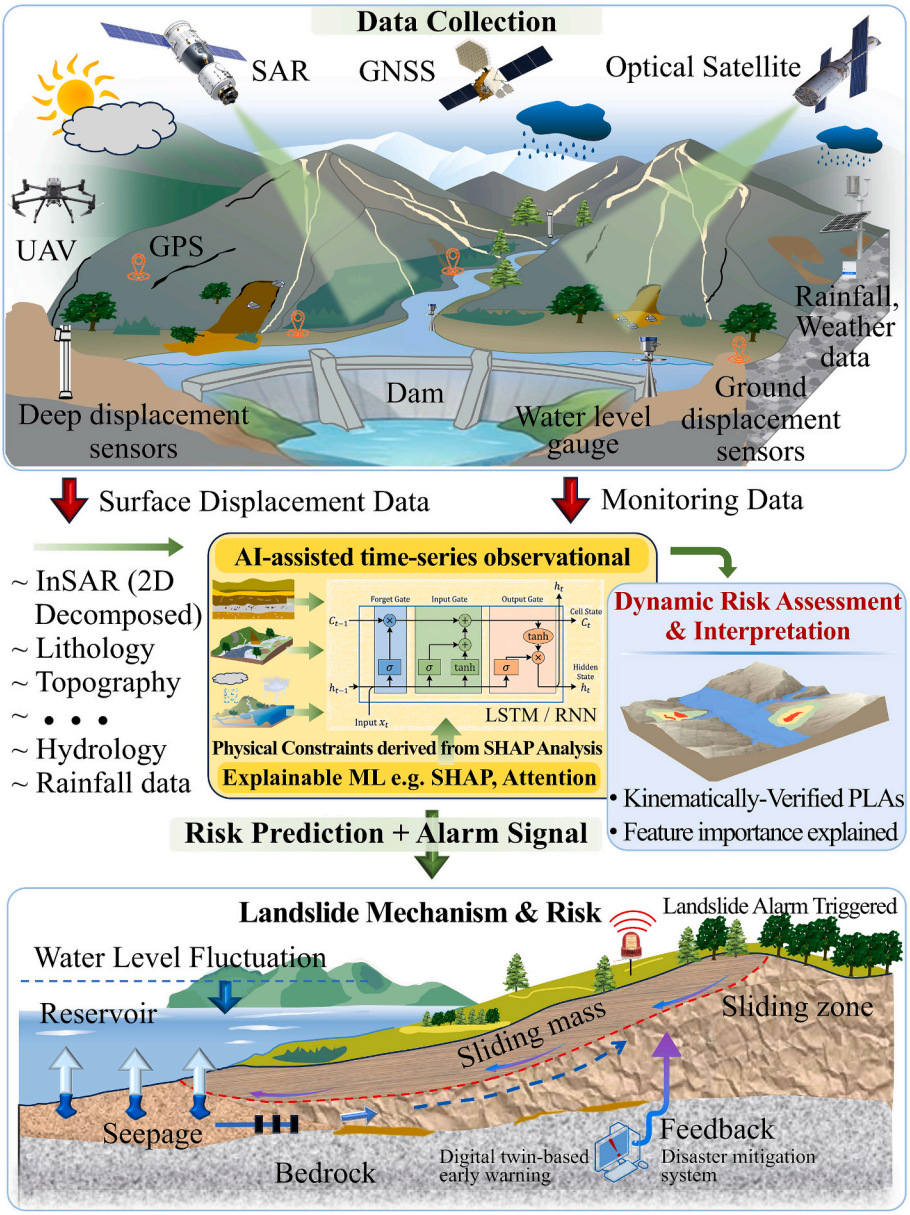

Fig. 22. Integrating dynamic monitoring and triggering mechanisms for landslide failure prediction.

and misclassify low-velocity creep under skewed InSAR velocity distributions, the Jenks Natural Breaks method better preserves natural groupings in the deformation data by minimizing intra-class variance. As a result, the derived thresholds (e.g., V5 < -34 mm/yr) are more consistent with the underlying deformation structure. In addition, the contingency matrix follows a conservative decision logic, assigning the highest hazard level only where very high susceptibility coincides with high deformation, which helps preserve the spatial coherence of high-risk zones under minor parameter adjustments.

Although the proposed approach increased the correctly identified hazardous area by a factor of four compared with rigid thresholding, resolution-related limitations remain (Fig. 21a and b). The 30 m grid resolution and strong spatial clustering may occasionally blur hazard boundaries or omit micro-landslides that fall below the interferometric detection limit. Future improvements incorporating high-resolution DEMs ($\leq$5 m), adaptive spatial kernels, and temporal deformation trajectories may help distinguish transient surface noise from persistent deep-seated instability, further improving the dynamic fidelity of PLA delineation.

Importantly, the proposed framework may be adaptable to other slope-hazard contexts, although this remains to be tested. Its main contribution lies in providing a transferable state-based decision strategy rather than a site-specific mapping rule. The framework can be adapted to other slope hazards (e.g., debris flows) by redefining the activity proxy. Similarly, for reservoir regions with different geological settings, the matrix boundaries can be adjusted according to local susceptibility classes and deformation characteristics without changing the underlying framework. This flexibility highlights the broader potential of the contingency matrix approach for hazard mapping and decision-support applications.

5.4. *Limitations and future outlook*

The primary advantage of the proposed framework lies in its state-based integration of susceptibility and deformation through a physically constrained contingency matrix. Refining negative samples ensures that the susceptibility baseline reflects genuine geological predisposition, leading to more reliable hazard zonation. However, the framework

remains subject to omission errors because of the inherent physical limitations of C-band interferometry. First, vegetation-induced decorrelation generates severe phase noise in densely vegetated gullies, masking small-scale failures (Wang et al., 2020). Second, kinematic ambiguity restricts the detection of movements oriented strictly parallel to the north-south axis, as well as slow creep rates below the interferometric detection threshold. Addressing these observational blind spots will require integrating L-band SAR missions (e.g., NISAR) and future high-resolution bistatic systems (e.g., TanDEM-L) to improve canopy penetration and better constrain multi-component displacement fields (Chapman et al., 2024).

Beyond these observational constraints, transitioning from static susceptibility zonation to dynamic failure forecasting represents a critical next step (Fig. 22). Because deformation processes in reservoir environments are driven by transient hydro-mechanical forcing, the physical drivers of instability inevitably evolve over time. While the current SHAP analysis provides an interpretable estimate of spatial predisposition, it provides only a static snapshot of feature importance. Explaining model behavior for each evolving state remains an advanced but valuable research direction.

To address this challenge, future research could incorporate time-series forecasting architectures, such as Long Short-Term Memory (LSTM) networks or Gaussian Processes (GPs) (Nava et al., 2023). Such integration would enable quantification of feature importance across distinct temporal states, helping to clarify, for instance, how the influence of Dis_River intensifies during rapid drawdown phases while rainfall-related controls become secondary. By embedding these temporally interpretable models into real-time monitoring and digital twin systems, future studies may improve understanding of failure mechanisms and support the development of precise, state-specific early warning criteria. In this context, the static susceptibility-deformation coupling presented in this study provides a reliable spatial prior for future dynamic forecasting frameworks.

6. Conclusion

This study establishes an interpretable framework for state-based landslide assessment in reservoir environments. Validated in the Baihetan Reservoir, the framework couples static environmental susceptibility with dynamic deformation through a contingency matrix, integrating optimized negative sampling, dual-orbit 2D InSAR kinematics, and explainable ML.

The results highlight three main findings. First, the reliability of negative sample selection plays an important role in susceptibility modeling. The IVM-based optimization strategy effectively reduces noise introduced by misclassified stable samples and improves model performance in deformation-prone reservoir terrains. Second, the contingency matrix approach provides a more robust means of combining susceptibility predictions with InSAR-derived deformation than conventional threshold-based methods. By representing susceptibility and deformation as coupled states rather than binary conditions, the framework improves the delineation of PLAs while reducing false detections associated with anthropogenic surface deformation. Third, SHAP-based model interpretation indicates that proximity to the reservoir and related hydrological processes are strongly associated with landslide susceptibility in the study area, emphasizing the role of hydro-mechanical forcing within the reservoir fluctuation zone.

Overall, this study suggests that integrating multi-geometry InSAR observations with interpretable ML can improve the identification of landslide-prone areas in reservoir environments. More fundamentally, susceptibility and deformation should be treated as complementary indicators of spatial predisposition and kinematic state rather than as interchangeable outputs. This distinction provides a clearer basis for developing more interpretable remote-sensing frameworks for landslide hazard assessment in evolving reservoir environments.

Some limitations remain. InSAR measurements derived from C-band Sentinel-1 data may be affected by decorrelation in densely vegetated terrain and show limited sensitivity to north-south displacement components due to satellite viewing geometry. Future work could integrate longer-wavelength SAR missions, such as L-band systems, and incorporate complementary ground-based monitoring data to further improve deformation detection and hazard characterization.

CRediT authorship contribution statement

Ke Xing: Visualization, Software, Methodology, Investigation, Conceptualization, Writing – review & editing, Writing – original draft. **Suju Li:** Validation, Supervision, Resources, Investigation, Data curation, Conceptualization, Writing – review & editing. **Jie Dou:** Visualization, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization, Writing – review & editing. **Fang Chen:** Validation, Supervision, Project administration, Investigation, Data curation. **Lizhe Wang:** Validation, Supervision, Software, Resources, Formal analysis, Conceptualization. **Aonan Dong:** Visualization, Software, Methodology, Formal analysis. **Hamza Daud:** Validation, Software, Methodology, Conceptualization. **Lele Zhang:** Visualization, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 42477170 and No. 42090054), the Natural Science Foundation of Hubei Province of China (No. 2025AFD435, No. 2022CFA002, and No. 2024AFD358). The Sentinel-1 datasets were freely provided by the European Space Agency (ESA).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2026.115442.

Data availability

Sentinel-1 SAR data are publicly available from the Alaska Satellite Facility. The Copernicus DEM is available from the Copernicus Data Space Ecosystem. Geological, hydrological, road infrastructure, and meteorological datasets were obtained from the corresponding official data providers as described in Table 1. The landslide inventory and field investigation data are available from the corresponding author upon reasonable request, subject to data-sharing restrictions.

References

Agliardi, F., Scuderi, M.M., Fusi, N., Collettini, C., 2020. Slow-to-fast transition of giant creeping rockslides modulated by undrained loading in basal shear zones. Nat. Commun. 11, 1–11. https://doi.org/10.1038/s41467-020-15093-3.

Al-Najjar, H., Kalantar, B., Pradhan, B., Beydoun, G., Ueda, N., 2025. SHapley Additive exPlanations (SHAP) for landslide susceptibility models: shedding light on explainable AI. ISPRS annals of the photogrammetry. Remote Sens. Spatial Inform. Sci. 81–85. https://doi.org/10.5194/isprs-annals-X-G-2025-81-2025. X-G-2025.

Berardino, P., Fornaro, G., Lanari, R., Sansosti, E., 2002. A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms. IEEE Trans. Geosci. Remote Sens. 40, 2375–2383. https://doi.org/10.1109/TGRS.2002.803792.

Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32. https://doi.org/10.1023/A:1010933404324.

Castelvecchi, D., 2016. Can we open the black box of AI? Nature 538, 20–23. https://doi.org/10.1038/538020a.

Chapman, B., Anconitano, G., Borsa, A., Christensen, A., Cushman, K.C., Das, A., Donnellan, A., Downs, B., Fielding, E., Holt, B., Joughin, I., Kellndorfer, J., Kim, S.,

McDonald, K., Meyer, F.J., Oliver-Cabrera, T., Parra, A., Patnai, C., Peacock, A.M., Pinto, N., Putrevu, D., Rosen, P., Saatchi, S., Simons, M., Siqueira, P., Taglialatela, C., Tymofyeyeva, E., Vaccaro, A., Zinke, R., Zwieback, S., 2024. The NASA ISRO SAR (NISAR) mission - validation of science measurement requirements. In: International Geoscience and Remote Sensing Symposium (IGARSS). Institute of Electrical and Electronics Engineers Inc, pp. 6623–6627. https://doi.org/10.1109/IGARSS53475.2024.10640858.

Chen, L., Ma, P., Yu, C., Zheng, Y., Zhu, Q., Ding, Y., 2023. Landslide susceptibility assessment in multiple urban slope settings with a landslide inventory augmented by InSAR techniques. Eng. Geol. 327. https://doi.org/10.1016/j.enggeo.2023.107342.

Ciampalini, A., Raspini, F., Lagomarsino, D., Catani, F., Casagli, N., 2016. Landslide susceptibility map refinement using PSInSAR data. Remote Sens. Environ. 184, 302–315. https://doi.org/10.1016/J.RSE.2016.07.018.

Dai, K., Chen, C., Shi, X., Wu, M., Feng, W., Xu, Q., Liang, R., Zhuo, G., Li, Z., 2023. Dynamic landslides susceptibility evaluation in baihetan dam area during extensive impoundment by integrating geological model and InSAR observations. Int. J. Appl. Earth Obs. Geoinf. 116, 103157. https://doi.org/10.1016/j.jag.2022.103157.

Daud, H., Dou, J., Khan, N.G., Xu, B., Dong, S., Dong, A., Ma, H., 2025. Tree-based machine learning and flow simulation for debris flow susceptibility, runout propagation, and dynamics in the higher Himalayas. Math. Geosci. 57 (7), 1195–1233. https://doi.org/10.1007/S11004-025-10196-3.

Daviran, M., Shamekhi, M., Ghezelbash, R., Maghsoudi, A., 2023. Landslide susceptibility prediction using artificial neural networks, SVMs and random forest: hyperparameters tuning by genetic optimization algorithm. Int. J. Environ. Sci. Technol. 20, 259–276. https://doi.org/10.1007/S13762-022-04491-3.

Dong, A., Dou, J., Fu, Y., Zhang, R., Xing, K., 2023. Unraveling the evolution of landslide susceptibility: a systematic review of 30-years of strategic themes and trends. Geocarto Int. 38. https://doi.org/10.1080/10106049.2023.2256308.

Dou, J., Yunus, A.P., Bui, D.T., Merghadi, A., Sahana, M., Zhu, Z., Chen, C.W., Han, Z., Pham, B.T., 2020. Improved landslide assessment using support vector machine with bagging, boosting, and stacking ensemble machine learning framework in a mountainous watershed, Japan. Landslides 17, 641–658. https://doi.org/10.1007/s10346-019-01286-5.

Du, J., Song, C., Li, Zhenhong, Tomás, R., Li, Zheng, 2025. Kinematic Behavior and Sliding Geometry of Large Anthropogenic-induced Landslides Using Three-dimensional Time Series InSAR: Insights from the Li-Kan Road Landslide, 22, pp. 3319–3333. https://doi.org/10.1007/s10346-025-02542-7.

Friedman, J.H., 2001. Greedy function approximation: a gradient boosting machine. Ann. Stat. 29, 1189–1232. https://doi.org/10.1214/aos/1013203451.

Guo, Z., Tian, B., Zhu, Y., He, J., Zhang, T., 2024. How do the landslide and non-landslide sampling strategies impact landslide susceptibility assessment? — A catchment-scale case study from China. J. Rock Mech. Geotech. Eng. 16, 877–894. https://doi.org/10.1016/j.jrmge.2023.07.026.

Herrera, G., Gutiérrez, F., García-Davalillo, J.C., Guerrero, J., Notti, D., Galve, J.P., Fernández-Merodo, J.A., Cooksley, G., 2013. Multi-sensor advanced DInSAR monitoring of very slow landslides: the Tena Valley case study (Central Spanish pyrenees). Remote Sens. Environ. 128, 31–43. https://doi.org/10.1016/j.rse.2012.09.020.

Hu, X., Bürgmann, R., Schulz, W.H., Fielding, E.J., 2020. Four-dimensional surface motions of the Slumgullion landslide and quantification of hydrometeorological forcing. Nat. Commun. 11, 1–9. https://doi.org/10.1038/s41467-020-16617-7.

Huang, R., Pei, X., Fan, X., Zhang, W., Li, S., Li, B., 2012. The characteristics and failure mechanism of the largest landslide triggered by the Wenchuan earthquake, May 12, 2008, China. Landslides 9, 131–142. https://doi.org/10.1007/s10346-011-0276-6.

Kang, Y., Lu, Z., Zhao, C., Xu, Y., Kim, J. wo, Gallegos, A.J., 2021. InSAR monitoring of creeping landslides in mountainous regions: a case study in Eldorado National Forest, California. Remote Sens. Environ. 258, 112400. https://doi.org/10.1016/J.RSE.2021.112400.

Li, G.K., Moon, S., 2021. Topographic stress control on bedrock landslide size. Nat. Geosci. 14, 307–313. https://doi.org/10.1038/s41561-021-00739-8.

Liu, Y., Yao, X., Gu, Z., Li, R., Zhou, Z., Liu, X., Jiang, S., Yao, C., Wei, S., 2024. Research on automatic recognition of active landslides using InSAR deformation under digital morphology: a case study of the Baihetan reservoir, China. Remote Sens. Environ. 304. https://doi.org/10.1016/J.RSE.2024.114029.

Ma, P., Chen, L., Yu, C., Zhu, Q., Ding, Y., Wu, Z., Li, H., Tian, C., Fan, X., 2025. Dynamic landslide susceptibility mapping over last three decades to uncover variations in landslide causation in subtropical urban mountainous areas. Remote Sens. Environ. 326, 114800. https://doi.org/10.1016/J.RSE.2025.114800.

Marmoni, G.M., Antonielli, B., Caprari, P., Di Renzo, M.E., Marini, R., Mastrantoni, G., Mazzanti, P., Patelli, D., Bozzano, F., 2025. Semi-automatic ranking of landslide candidate areas at a regional scale using EGMS InSAR data for territorial planning and risk management. Landslides 22, 4077–4095. https://doi.org/10.1007/s10346-025-02597-6.

Nava, L., Carraro, E., Reyes-Carmona, C., Puliero, S., Bhuyan, K., Rosi, A., Monserrat, O., Floris, M., Meena, S.R., Galve, J.P., Catani, F., 2023. Landslide displacement forecasting using deep learning and monitoring data across selected sites. Landslides 20, 2111–2129. https://doi.org/10.1007/s10346-023-02104-9.

Özcan, Z., Caglayan, İ., Kabak, Ö., Kılıç Gül, F., 2025. Integrated risk mapping for forest fire management using the analytical hierarchy process and ordered weighted average: a case study in southern Turkey. Nat. Hazards 121, 959–1001. https://doi.org/10.1007/s11069-024-06810-y.

Qi, S., Xu, Q., Lan, H., Zhang, B., Liu, J., 2010. Spatial distribution analysis of landslides triggered by 2008.5.12 Wenchuan Earthquake, China. Eng. Geol. 116, 95–108. https://doi.org/10.1016/j.enggeo.2010.07.011.

Reichenbach, P., Rossi, M., Malamud, B.D., Mihir, M., Guzzetti, F., 2018. A review of statistically-based landslide susceptibility models. Earth Sci. Rev. 180, 60–91. https://doi.org/10.1016/j.earscirev.2018.03.001.

Sharifi, S., Macciotta, R., Hendry, M., Rotheram-Clarke, D., Huntley, D., 2024. Evaluating topography-based methods in 3D decomposition of InSAR 1D velocities obtained for translational landslides: Thompson River valley in Canada. Landslides 21, 411–427. https://doi.org/10.1007/S10346-023-02153-0.

Song, Y., Hu, X., Shi, X., Cui, Y., Zhou, C., Xu, Y., 2025. Hydrological proxy derived from InSAR coherence in landslide characterization. Remote Sens. Environ. 322. https://doi.org/10.1016/j.rse.2025.114712.

Tang, H., Wasowski, J., Juang, C.H., 2019. Geohazards in the Three Gorges Reservoir Area, China – lessons learned from decades of research. Eng. Geol. https://doi.org/10.1016/j.enggeo.2019.105267.

Tian, F., Zhang, W., Zhu, H.H., Wang, C., Chang, F.N., Li, H.Z., Tan, D.Y., 2025. Multi-temporal InSAR-based landslide dynamic susceptibility mapping of Fengjie County, Three Gorges Reservoir Area, China. J. Rock Mech. Geotech. Eng. https://doi.org/10.1016/j.jrmge.2025.01.012.

Wang, L., Marzahn, P., Bernier, M., Ludwig, R., 2020. Sentinel-1 InSAR measurements of deformation over discontinuous permafrost terrain, northern Quebec, Canada. Remote Sens. Environ. 248. https://doi.org/10.1016/j.rse.2020.111965.

Wang, X., Crosta, G.B., Clague, J.J., Stead, D., Sun, J., Qi, S., Liu, H., 2021. Fault controls on spatial variation of fracture density and rock mass strength within the Yarlung Tsangpo Fault damage zone (southeastern Tibet). Eng. Geol. 291. https://doi.org/10.1016/j.enggeo.2021.106238.

Wang, Y., Wang, L., Liu, S., Han, L., Zhang, W., Hong, L., Zhu, Z., Zhu, X., 2025. Region similarity assessment for empowering physics-informed transfer learning-based landslide susceptibility mapping. J. Rock Mech. Geotech. Eng. https://doi.org/10.1016/j.jrmge.2025.06.030.

Yan, Y., Wang, Y., 2021. Determining suitable spaceborne SAR observations and ground control points for surface deformation study in rugged terrain with InSAR technique. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 14, 11324–11334. https://doi.org/10.1109/JSTARS.2021.3123326.

Yang, S., Li, D., Liu, Y., Xu, Z., Sun, Y., She, X., 2023. Landslide identification in human-modified alpine and canyon area of the Niulan river basin based on SBAS-InSAR and optical images. Remote Sens. https://doi.org/10.3390/rs15081998.

Yao, J., Yao, X., Zhao, Z., Liu, X., 2023. Performance comparison of landslide susceptibility mapping under multiple machine-learning based models considering InSAR deformation: a case study of the upper Jinsha River. Geomat. Nat. Haz. Risk 14. https://doi.org/10.1080/19475705.2023.2212833.

Yao, J., Wang, Y., Wang, T., Zhang, B., Wu, Y., Yao, X., Zhao, Z., Zhu, S., 2024. Assessing geological hazard susceptibility and impacts of climate factors in the eastern Himalayan syntaxis region. Landslides 21, 2195–2209. https://doi.org/10.1007/s10346-024-02262-4.

Yao, J., Wang, T., Yao, X., 2025. Spatial-temporal evolution of landslides spanning the impoundment of Baihetan mega hydropower project revealed by satellite radar interferometry. Remote Sens. Environ. 321, 114668. https://doi.org/10.1016/J.RSE.2025.114668.

Yin, Y., Huang, B., Wang, W., Wei, Y., Ma, X., Ma, F., Zhao, C., 2016. Reservoir-induced landslides and risk control in Three Gorges Project on Yangtze River, China. J. Rock Mech. Geotech. Eng. 8, 577–595. https://doi.org/10.1016/j.jrmge.2016.08.001.

Zhang, L., Zhang, R., Dou, J., Hou, S., Xiang, Z., Wang, H., Yang, P., Liu, X., 2024a. Advancing reservoir landslide stability assessment via TS-InSAR and airborne LiDAR observations in the Daping landslide group, Three Gorges Reservoir Area, China. Landslides. https://doi.org/10.1007/s10346-024-02337-2.

Zhang, R., Zhao, X., Dong, X., Dai, K., Deng, J., Zhuo, G., Yu, B., Wu, T., Xiang, J., 2024b. Potential landslide identification in baihetan reservoir area based on C-/L-band synthetic aperture radar data and applicability analysis. Remote Sens. (Basel) 16. https://doi.org/10.3390/rs16091591.

Zheng, D., Li, Y., Yan, C., Wu, H., Yamashiki, Y.A., Gao, B., Nian, T., 2025. Landslide susceptibility assessment using AutoML-SHAP method in the southern foothills of Changbai Mountain, China. Landslides 22, 1855–1875. https://doi.org/10.1007/s10346-025-02462-6.